Coffeehouses and the Rise of Science

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[Most recent version here]

Motivation

Producing new knowledge depends on **access to the existing knowledge stock**

Much of existing knowledge **written down, cheap to access**

At the **frontier** and in periods of rapid change, largely **tacit**: access through people

- Who participates; who meets whom; where people circulate

Universities **filter access**: quality and coordination, but a **narrower pool** of talent

Firms and cities **pay to lower the cost of access**: return-to-office mandates, science parks

Does broader access to tacit knowledge accelerate knowledge production?

Motivation

New social institutions of the 17th and 18th centuries as a **test bed**

- English coffeehouses, French salons, German table societies

Coffeehouses as the prime example: **public**, discursive, open

> *The coffee house not merely made **access to the relevant circles less formal and easier**; it **embraced the wider strata of the middle class, including craftsmen** and shopkeepers.*
>
> Jürgen Habermas, *The Structural Transformation of the Public Sphere*, 1962 [1989]

Mokyr (2005): practical men meeting scholars, cheaper "useful knowledge"

Did coffeehouses accelerate applications of science, and, if so, why?

This Paper

Studies coffeehouses and applied science in 17th-century London

- Center of the Scientific Revolution; one of Europe's first coffeehouses, 1652

Builds a comprehensive microdataset on 17th-century London

- Every recorded coffeehouse and applied scientist; 10+ other geocoded variables

Estimates the effect of staggered coffeehouse entry on applied science across parishes

- Not explained by pre-trends, local commercial booms, or record survival

Unpacks some mechanisms

Finds supporting evidence in text data and across European cities

Quantifies the effect on London as a whole

- Urban model calibrated to the parish and cross-city estimates

Preview of Results

Coffeehouses raise applied science. After a parish's first one, a persistent rise:

- Mathematical practitioners roughly double (from 0.62)
- Scientific instrument shops nearly fivefold (from 0.33)

Why? Broader access to exchange beyond academic networks

- Effects concentrated among those with **weak family ties to Oxford and Cambridge**
- More **connections linking craft skill to theory**
- New shops turn toward **experimental instruments**

Prominent scientist Hooke's diary: the mechanisms at work

Across European cities: titles and instrument makers rise after the first coffeehouse

London as a whole: 31% of applied scientists, 1650s–1690s, due to coffeehouses

Contribution to the Literature

Cities raise invention when proximity becomes contact (Andrews, 2026; Arzaghi and Henderson, 2008; Atkin, Chen, and Popov, 2022; Glaeser, 1999)

- **Long-run** effect of an institution turning proximity into **broadened exchange in one city**

Cheaper access to codified knowledge increases growth and innovation (Berkes and Nencka, 2026; Dittmar, 2011; Juhász, Sakabe, and Weinstein, 2026)

- Causal estimate for **an institution that broadened access to tacit knowledge**

Growth in England begins in the early 1600s, progress-oriented language in the later 1600s (Almelhem et al., 2026; Bouscasse, Nakamura, and Steinsson, 2025; Kelly and Ó Gráda, 2022)

- **A social institution that arrived in London in 1652**, raising applied science where it opened

Historians credit coffeehouses with fostering science (Cowan, 2005; Stewart, 1999)

- First causal evidence: coffeehouses raised applied science and **broadened participation**

Outline

Background

Data

Effects on Science in London

Mechanisms

Extension to Europe

Aggregate Implications for London

Conclusion

Background

London in the 17th Century

Population more than doubled: 200,000 (1600) to 475,000 (1670)

Capital, trade and manufacturing hub; **no university**

Londoners with cash for print, news, exotic imports

Social life in taverns, shops, playhouses, the Exchange

VISSCHER'S VIEW OF LONDON, 1616

A Divided Scientific Community

By 1600: small scientific community; **tacit** knowledge

Useful to practitioners and specialized merchants:

- Mathematical practitioners e.g., surveyors, teachers, instrument makers
- Scientific instrument shops e.g., for measurement, navigation, observation

Practitioners **divided by status**: university training outranked self-teaching

- Rare contact, through a few bridges, mainly instrument shops

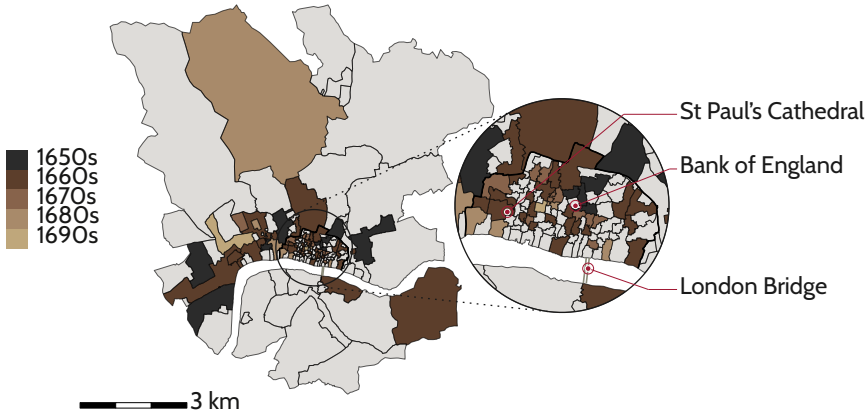
Enter the Coffeehouse

London's first coffeehouse, 1652, backed by a Levant Company merchant

Merchants followed, for curiosity and commerce: 82 coffeehouses by 1663

Entry for a penny, about a tenth of a day's wage: **open to men of any social rank**

Mapping the First Coffeehouses in London



Coffeehouse Talk in Hooke's Diary Is Discursive

3,291 coffeehouse passages · 1672–1683 · size: log-odds against 15,102 phrases naming no public house

Coffeehouses, Science, and Access

Historians: coffeehouses fostered scientific exchange

- Experimental philosophers met and traded with instrument makers
- Experimental ideas debated in public

Contemporaries: coffeehouses **broadened access to tacit knowledge**

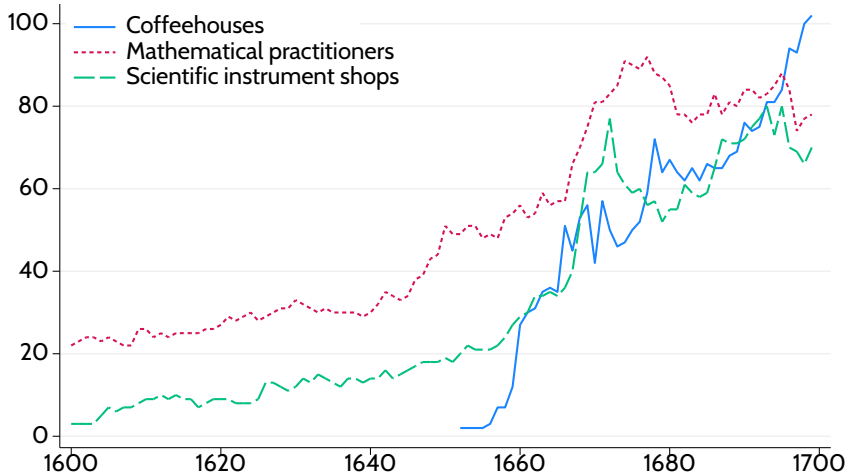
> ***Coffee-houses make all sorts of people sociable, the rich and the poor meet together, as also do the learned and unlearned**: It improves arts, merchandize, and all other knowledge; for here an inquisitive man […] **may get more in an evening than he shall by Books in a month**.*
>
> John Houghton, Fellow of the Royal Society, 1701

Clockmaker Thomas Tompion (1639–1713) in Hooke's Diary

1639	Baptized in Northill, Bedfordshire, the **son of a blacksmith**.
1671	Joins the **Clockmakers' Company** in London with **no scholarly standing**.
27 June 1674	"**Told him at Childs coffee house of new Dividing compasses** screw upon a Rule."
8 March 1675	"**At Garaways [coffee house]. Tompion. I shewd my way of fixing Double Springs** to the inside of the Ballance wheel."
7 April 1675	"**With the King and shewd him my new spring watch, Sir J. More and Tompion there** […] commended it far beyond Zulichems."
1676	Makes **two year-going clocks**, commissioned by Sir Jonas Moore, **for the new Royal Observatory** at Greenwich.

Coffeehouse meetings joined craft skill to experimental science

Coffeehouses and Applications of Science



Data

Dataset

Coffeehouses

Mathematical practitioners

Scientific instrument shops

Geocoding

Dataset

London panel

162 **parishes** by **decade**, 1600s–1690s

Geocoded annual address-level variables:

Treatment	Outcomes	10+ Covariates
Coffeehouses	Mathematical practitioners Scientific instrument shops	Economic, social, and demographic indicators

Hooke's diary (1672–1683)

3,464 diary days transcribed

European cities

200 **cities** by **decade**, 1600s–1690s, variables analogous to the London panel

Coffeehouses

286 coffeehouses (Lillywhite, 1963, 1972)

Names, addresses, and years of recorded mentions

Three-fifths of those counted in the 1690s, a third to a half of the 1663 survey; tracks print mentions Details

Coffeehouse Handbill 1652

The Turk's Head Coffeehouse Token 1669

Primary Sources

972. Pasqua Rosee's Coffee House, *St. Michael's Alley, Cornhill. Has the distinction of being the first coffee-house set-up in London; but its origin is sadly confused in early writings.*

1652–66 The earliest known advertisement for making and selling coffee in London is in the British Museum. No date is printed on the sheet which is headed: 'The Vertue of the COFFEE Drink. Firſt publiquely made and ſold in England by Paſqua Roſee. . . .' The sheet bears the following footnote: 'Made and Sold in St. Michael's Alley in Cornhill, by Paſqua Roſee at the Signe of his own Head.'

The buildings in St. Michael's Alley perished in the Fire September 1666 and the house of Pasqua Rosee came to an end.

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Mathematical Practitioners

244 mathematical practitioners (Taylor, 1954; Warren et al., 2016)

Practice mathematics in teaching, writing, instrument making, and technical work

- 1/3 in the *Oxford Dictionary of National Biography*

Names, workplace addresses, years of "flourishing", and social network

Other practitioners	*N*	Teachers	*N*	Makers	*N*
All 25 distinct titles	69	All 13 distinct titles	54	All 20 distinct titles	130
Mathematical practitioner	28	Mathematics teacher	24	Instrument maker	46
Surveyor	10	Mathematics professor	6	Mathematical instrument maker	28
Hydrographer	4	Mathematics schoolmaster	6	Optical instrument maker	19
Land surveyor	4	Navigation teacher	4	Dial maker	6
Mathematician	4	Practical mathematics teacher	3	Globe maker	6

164. FORSTER, WILLIAM (fl. 1630–74), was the pupil and friend of William Oughtred (94) and published the description of his Circles of Proportion in which Delamain (122) was attacked. He *taught practical mathematics* and undertook surveying, while his friend William Leybourne (222) describes a protractor of his design. He also made almanacks and calendars, of which there was an example in the mathematical section of Ralph Thoresby's museum in 1714, besides one in the collection of the late Mr Gabb. When in London he lived at the Red Bull, 'over against St Clement's Churchyard, outside Temple Bar', but he also practised surveying in Ireland.

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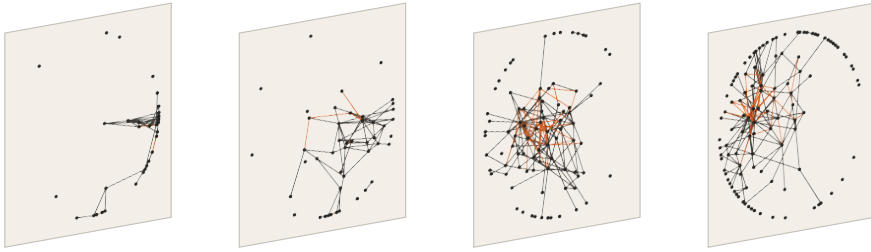
Mathematical Practitioners

Evolving Network of Recorded Social and Intellectual Ties



Ties in Taylor (1954) plus additional ties in Warren et al. (2016)

Scientific Instrument Shops

345 scientific instrument shops (Clifton, 1995; Hay et al., 2025)

Makers of scientific instruments

Masters, apprentices, addresses, and years of activity

Practical instruments

Sector 1623

Compendium 1630–1653

Experimental instruments

Microscope 1665

Barometer 1695–1705

QUARE Daniel
1671 d 1724
Clock M, Watch M
1680 King's Arms, Exchange Alley, off Lombard St, London
1715 Exchange Alley in Cornhill, London
Guild: Clockmakers, fr 1671
Partnership: QUARE & HORSEMAN*
Had apprentice:
DAVIS John (I)* 1685
HORSEMAN Stephen* 1701/2
13 others listed by Loomes but not known as IM
Succeeded by QUARE & HORSEMAN*
b 1649; Inv. portable barometer; Pat. barometer 1695; partnership 1709 onwards; Master CKC 1708
Known to have sold: barometer
Sources: GL: CKC 2710/3 fol. 28, 3939; Goodison (1977) p.206–20; King & Millburn (1978) p.124; Loomes (1981b)

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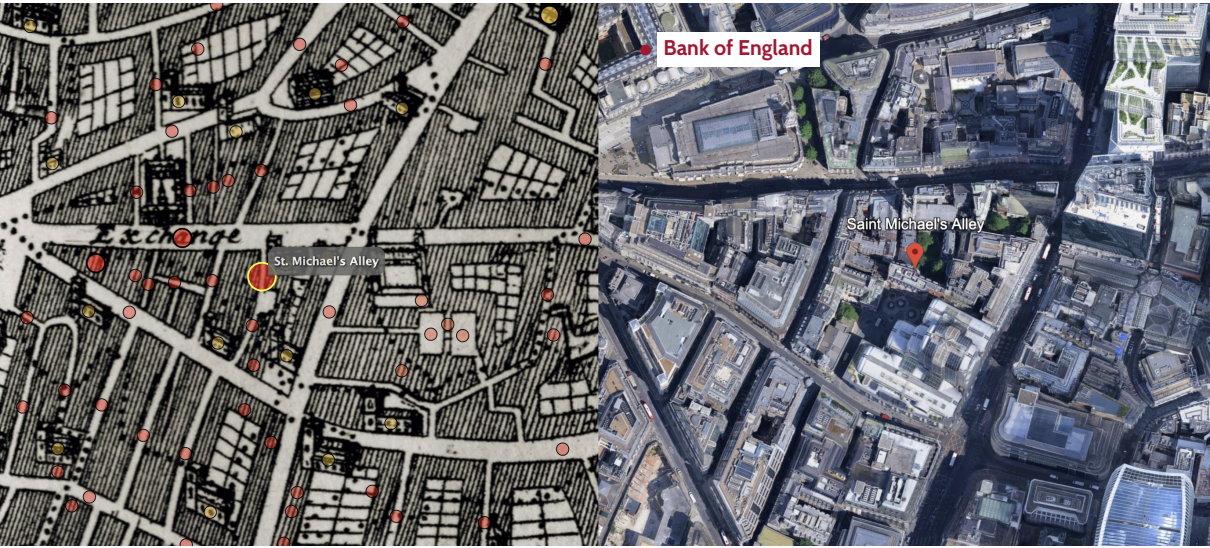
Geocoding

Addresses geocoded with Google Maps, historical maps and gazetteers

Coordinates mapped onto 1660 London parishes (Cummins, Kelly, and Ó Gráda, 2016)


Exchange
St. Michael's Alley
Bank of England
Saint Michael's Alley

Effects on Science in London

Identification

Empirical strategy

Results

Robustness checks

Taverns as a placebo treatment

Creation or relocation

Identification

Variation. Coffeehouse openings in 61 of 162 parishes, staggered from 1652

Assumptions. Absent entry, treated parishes track never-treated ones; no anticipation

- Plausibility tested: pre-trends, placebo outcomes, and more

Why ex ante plausible. Entry followed commerce, not science:

- Middle-class merchants selling coffee, sited for foot traffic Details
- Proprietors largely unconnected to practitioners before 1652 Details
- Clientele broad throughout the century Details

Empirical Strategy

Model: Staggered, heterogeneity-robust DiD across parishes p and decades t

$$y_{pt} = \alpha + \sum_{g=1650}^{1690} \sum_{\substack{s=1600 \\ s \neq g-10}}^{1690} \beta_{gs} \, \mathbb{1}[s = t] \, \mathbb{1}[g = E_p] + \gamma_p + \delta_t + \epsilon_{pt}$$

with

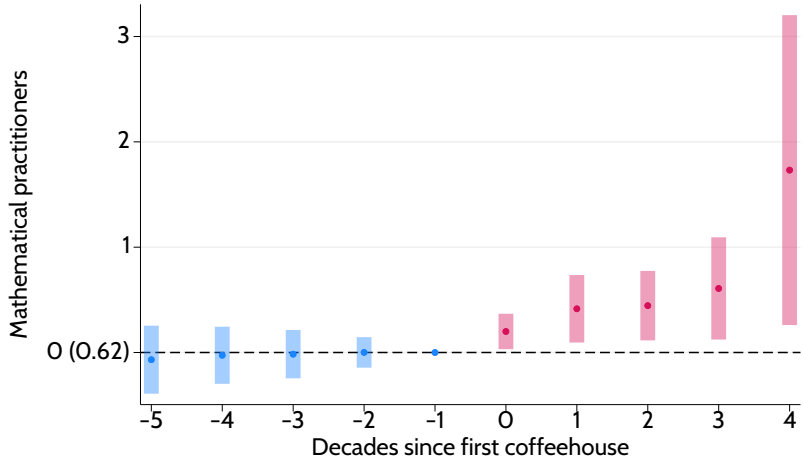
- y_{pt} = # math. practitioners (sci. instrument shops) in parish p, decade t
- E_p = decade of the 1st coffeehouse in parish p Details; γ_p, δ_t parish and decade FEs

Estimand: τ_h, treatment effect h decades after entry, relative to the decade before Details

Identification: Parallel trends and no anticipation

Estimation: Wooldridge (2023); never-treated controls; SEs clustered by parish

Coffeehouses Increase Scientific Activity



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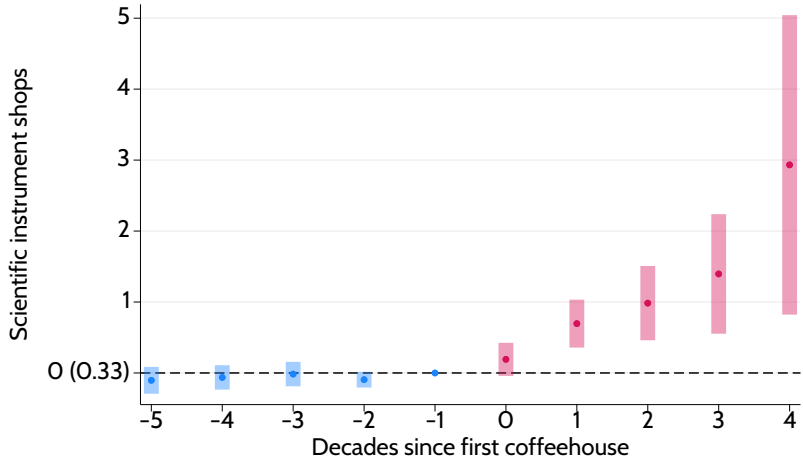


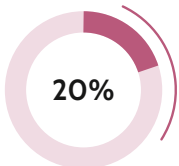
Coffeehouses Explain a Substantial Share of Scientific Activity

Effect per treated parish

Practitioners. 0.62 before entry, 1.30 after (2.1 times)

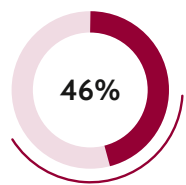
Instrument shops. 0.33 before entry, 1.57 after (4.8 times)

Share of activity accounted for by coffeehouse openings, 1650s–1690s



(A) Mathematical practitioners

502 practitioner-decades



(B) Scientific instrument shops

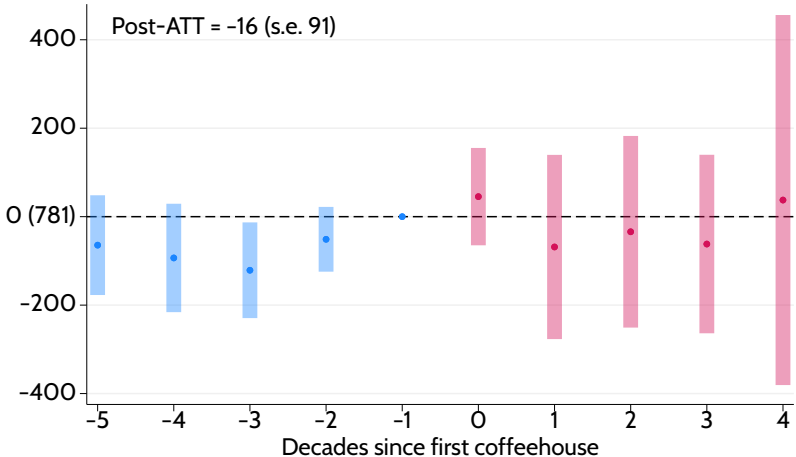
412 shop-decades

Robustness Checks

Placebo outcomes	Death density · Luxury shops · Commercial premises · Letters · Taverns · Baptism density
Record survival	Dropping sources · Records lost in the Fire · Guild shops · Roster overlap
Covariates	Balance · Balance in changes · IPW-adjustment · Regression-adjustment
Local confounders	Great Fire · London Wall
Randomization inference	Placebo coffeehouses · Placebo coffeehouses conditional on commerce
Treatment timing	2-, 3-, 5-year panels · Closures · Non-absorbing treatment · Treatment intensity
Spatial aggregation	Conley (1999) SEs · Hexagon cells & map
Trends and influence	Group-specific trends · HonestDiD · Leave one parish out
Estimators	IHS · Never- and not-yet-treated controls · Poisson

Effects on Placebo Outcomes

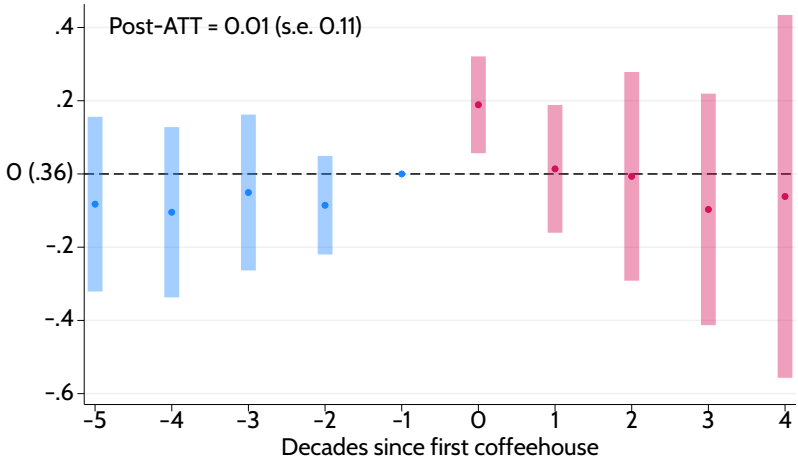
Death Density



Parishes with burial records: 116 of 162, 1600s–1690s

Effects on Placebo Outcomes

Luxury Shops



Composition

Taverns as a Placebo Treatment

Idea. More than the traditional sociability of taverns

Prediction. If meeting places alone mattered: similar effects from taverns

Tavern Talk in Hooke's Diary Is Convivial

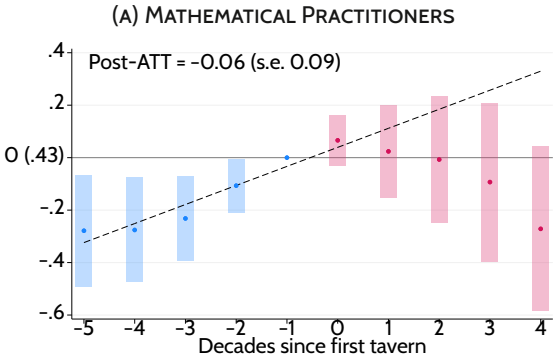
(a) A London Tavern, 1692

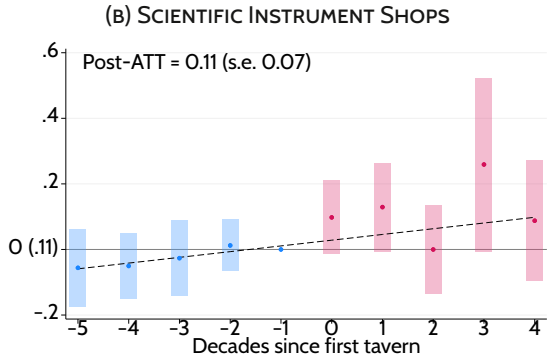
(b) Bigrams of Hooke's Tavern Visits, 1672–1683

Who a tavern day brings

261 tavern passages, 1672–1683 · size: log-odds vs. no public house

Taverns Show No Comparable Rise Before 1650





Subtract group-specific trends

Creation or Relocation?

Do coffeehouses **create or relocate** scientific activity?

Three patterns favor creation:

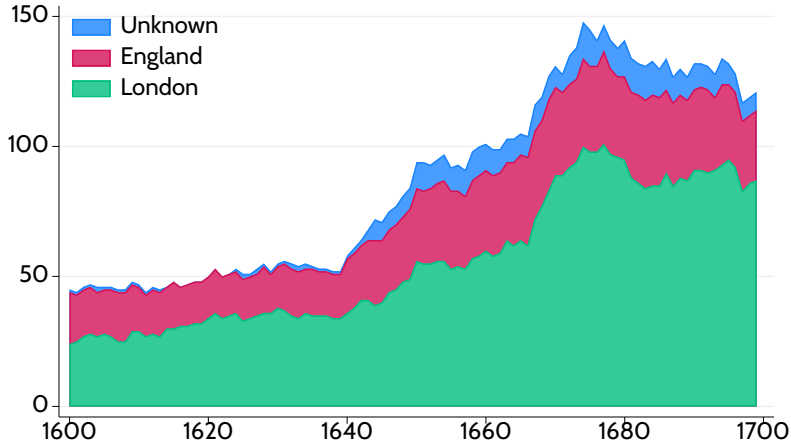
#1 Science grows in both London and England

#2 Spillovers onto neighboring cells, not losses

#3 Most of the local effect is new activity, not moves

#1 Science Grows in Both London and England

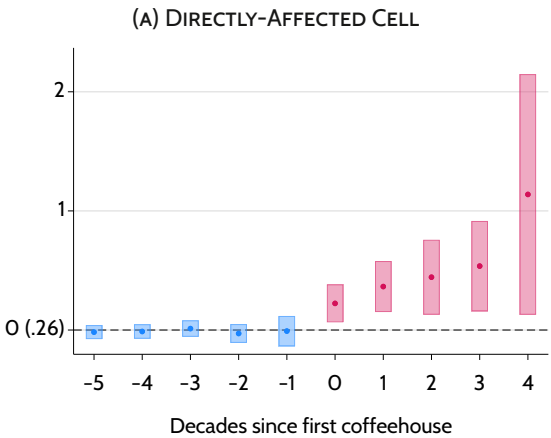
Mathematical Practitioners

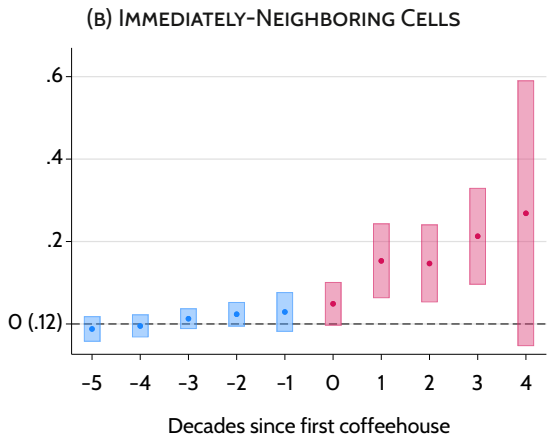


Scientific instrument makers

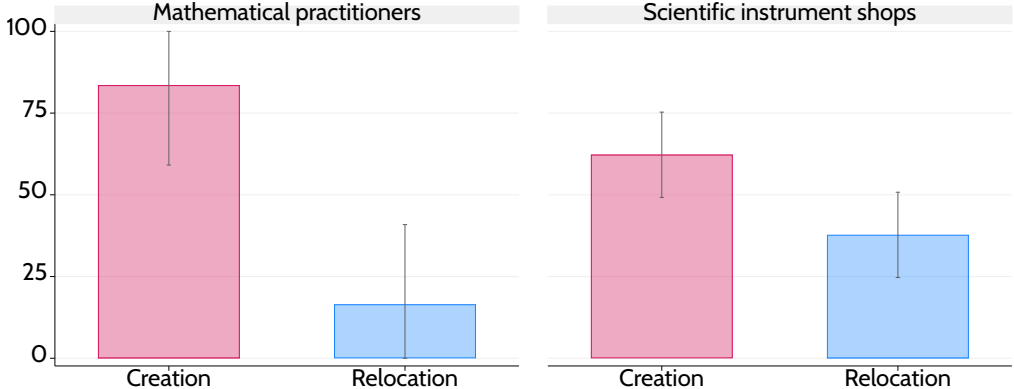
#2 Spillovers Are Inconsistent with Pure Relocation

Mathematical Practitioners





#3 Effects Not Driven by Relocation



Levels

Mechanisms

Barriers to entry

Connecting theory and practice

Experimental science

Putting practitioners into circulation

Barriers to Entry

Idea. Cheaper access for those without ties to the elite: **"penny universities"**

Prediction. Effects stronger for practitioners with weaker ties to elite institutions

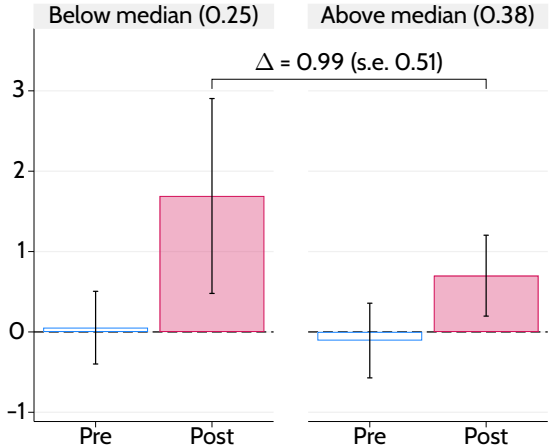
1. Proxy inherited ties by relative Oxbridge representation of surnames s (Clark and Cummins, 2014):

$$z(s) \equiv \frac{\text{share of } s \text{ among Oxbridge matriculants before 1650}}{\text{share of } s \text{ among English couples married 1538–1649}}$$

2. Re-estimate the baseline across partitions of $\{z(s) : s\}$

Effects Concentrated Outside Elite Networks

Below and Above Median Split



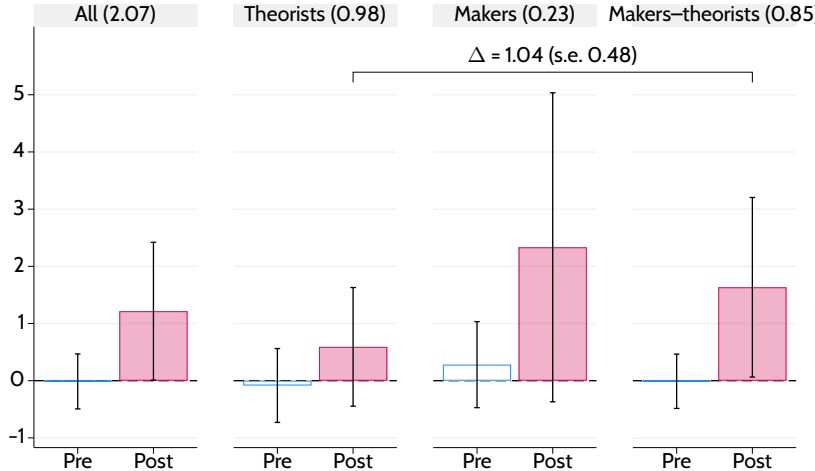
Along the Lorenz curve

Connecting Theory and Practice

Idea. Makers and theorists, previously apart, brought together

Prediction. More connections between theorists and makers

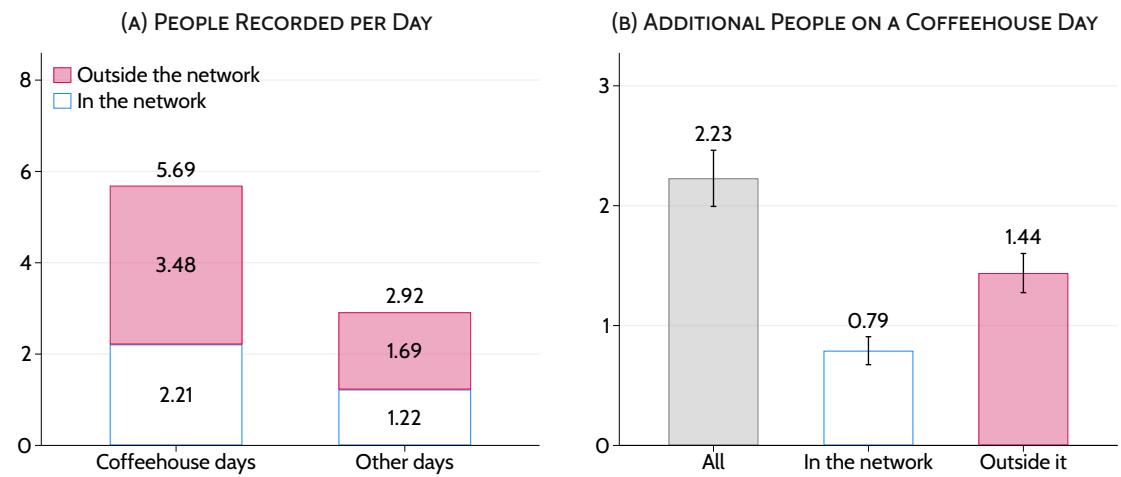
Coffeehouses Connected Makers with Theorists



All connection types

What changes hands

Robert Hooke's Recorded Company on Coffeehouse Days

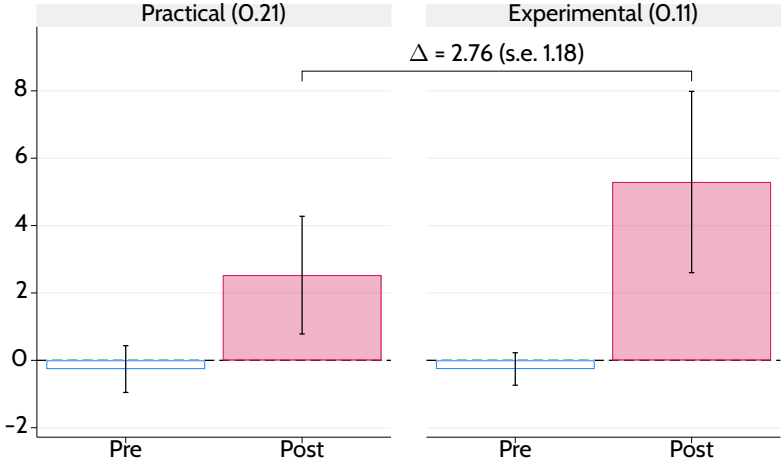


Experimental Science

Idea. Tacit knowledge of instruments and experiments changing hands in public

Prediction. More experimental instrument shops

Coffeehouses Advance Experimental Science

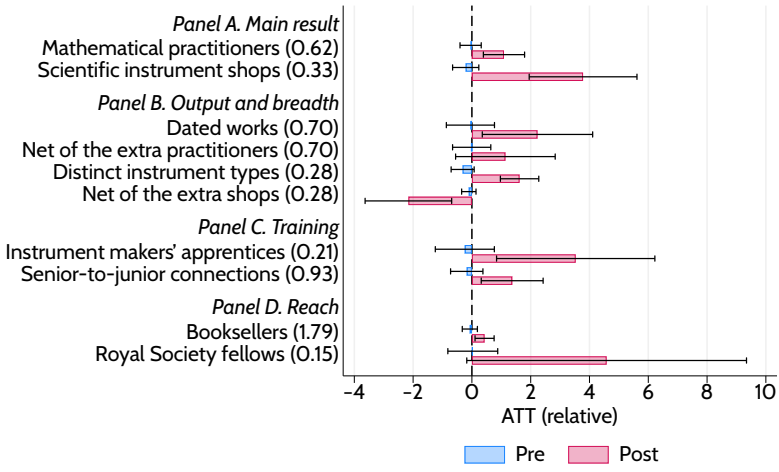


Putting Practitioners into Circulation

Idea. A wider circle of applied science

Prediction. A more general impact on applied science in coffeehouse parishes

Coffeehouses Raise Output, Training, and Reach



Extension to Europe

Coffeehouses as a Marker of Social Change in Europe

Does the London result scale up to other cities?

Coffeehouses **marked a broader shift toward open sociability and cheaper access**

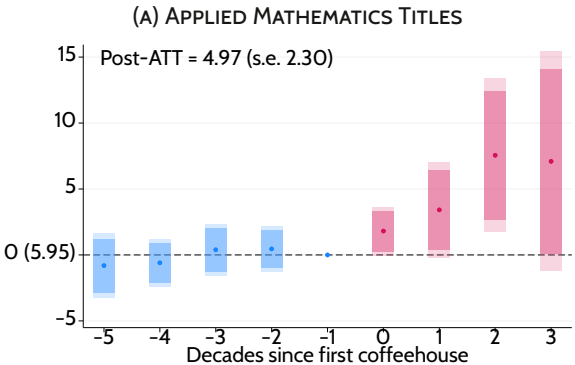
City-level **associations**, against cities of similar size and growth:

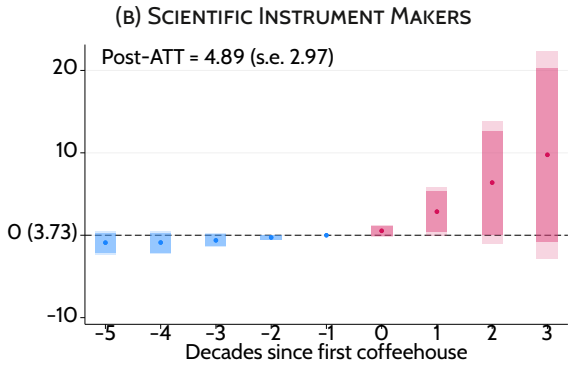
- Coffeehouses (own dataset from a systematic literature review) Details
- Applied mathematics titles (Universal Short Title Catalogue, 2026)
- Scientific instrument makers (R. S. Webster and M. K. Webster, 2007)

Mapping the First Coffeehouses in Europe

Coffee ports predict entry timing

Applied Science Moves with Coffeehouse Entry in Europe





Controls: population growth, 1500–1600, and log population in 1600, each interacted with decade FEs

Aggregate Implications for London

Why a Model?

Parish estimates: relative to other parishes, so creation or relocation?

City estimates: whole-city effects, but not where in the city

Model: combines both, to ask what London would look like without coffeehouses

A Coffeehouse as a Meeting Technology

Access: expected meetings with other scientists

- Coffeehouse in the parish: +74%; halves every 2.5 km (estimated, PPML)

Choices: two steps, Fréchet tastes

- Whether to do science: rises with London's average access (European cities)
- Where: toward parishes with more access (parish event study)

Without Coffeehouses, a Third Fewer Scientists

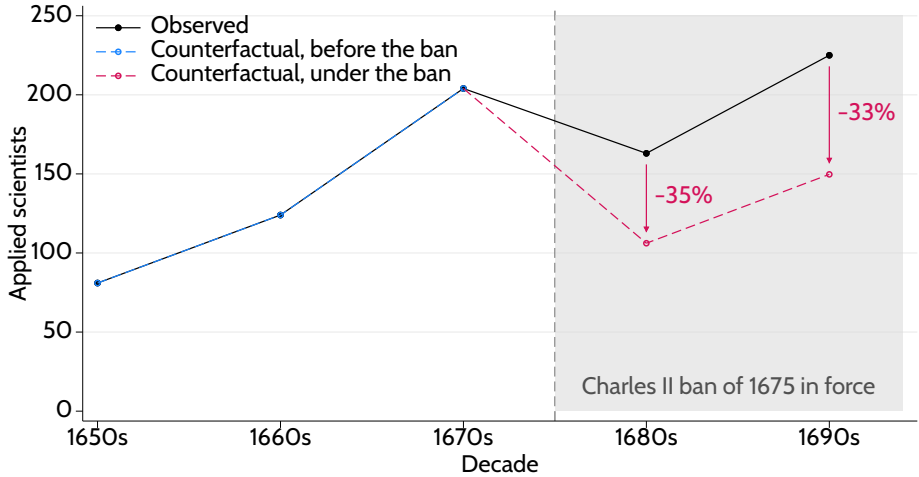
Remove every coffeehouse, 1650s–1690s

- London: 31% fewer applied scientists

Coffeehouse parishes' loss: mostly creation

- 77% leave science, 23% move to other parishes

Had Charles II's Ban of 1675 Held



Conclusion

Conclusion

Coffeehouses fell on the fertile ground of the Scientific Revolution, accelerating applied science

- 31% of London's applied scientists, 1650s–1690s, due to coffeehouses

Proximity turned into exchange across divided communities

- When experimental philosophy needed what craftsmen knew and books did not carry

The lever: cost of entry to a public sphere of knowledge exchange

- Workshops lowered it for those in the trade; coffeehouses, for a reserve of talent

Broader access to tacit knowledge can accelerate knowledge production

Where Do We Go from Here?

Where else did social venues widen who produces knowledge?

- Greek agora, Roman forum, salons, today's "third places"
- Consumer society breaking social barriers: other goods whose sale built places to meet?

What are today's coffeehouses?

- Coworking spaces, hackathons, science parks: who gains when access gets cheaper?
- As AI cheapens codified knowledge, do venues for tacit exchange matter more?

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Empirical Specification: Estimand

Dynamic effect h decades after entry, relative to the decade before:

$$\widehat{\tau}_h \equiv \frac{1}{|I_h|} \sum_{p \in I_h} \widehat{\tau}_{p,E_p+h}$$

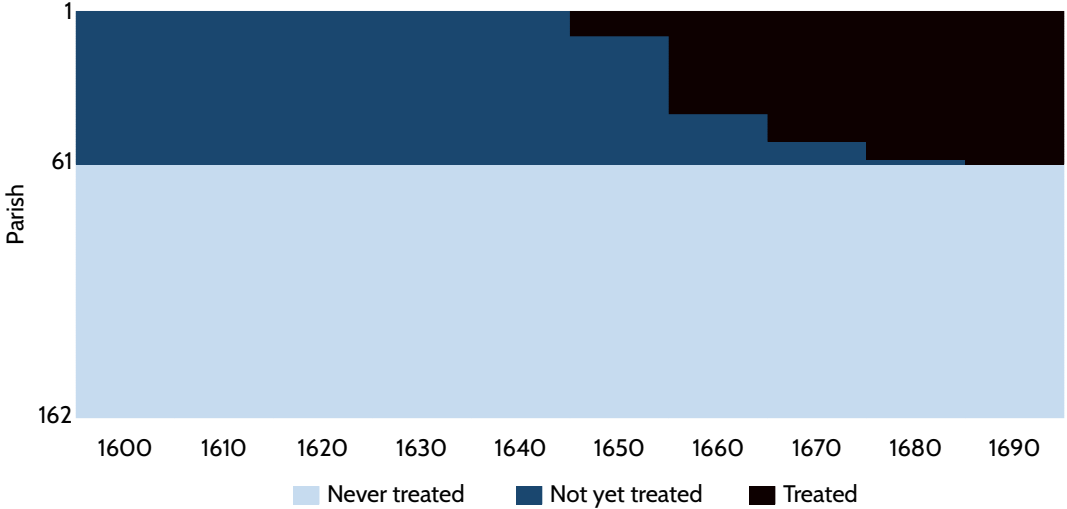
- I_h: treated parishes observed at $E_p + h$; $\widehat{\tau}_4$ from the 10 parishes treated in the 1650s

Imputation (Wooldridge, 2023): $\widehat{y}_{pt}(\infty)$ fitted from intercept and fixed effects alone

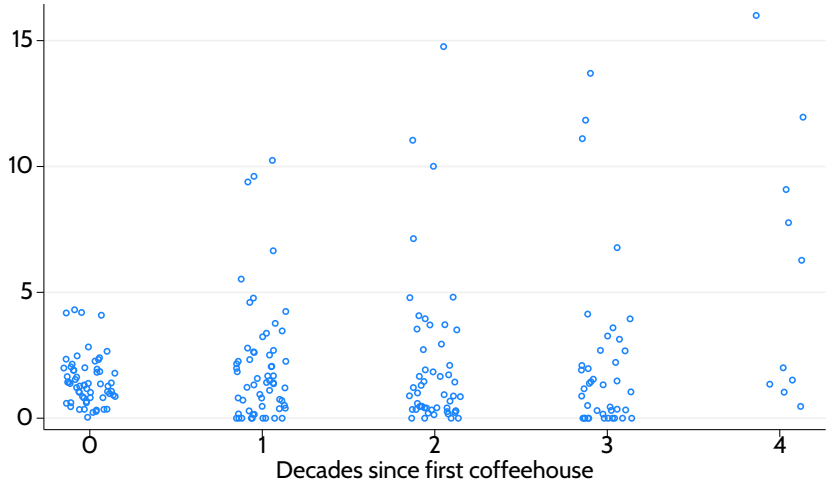
Parish-decade effect $\widehat{\tau}_{pt} \equiv \widehat{y}_{pt} - \widehat{y}_{pt}(\infty)$, with $\widehat{y}_{pt}$ from the full model

Equals the long differences of Callaway and Sant'Anna (2021); extends to Poisson

Treatment Rollout: First Coffeehouses



Coffeehouses Over Decades Since First Coffeehouse



Data Validation

Coffeehouses

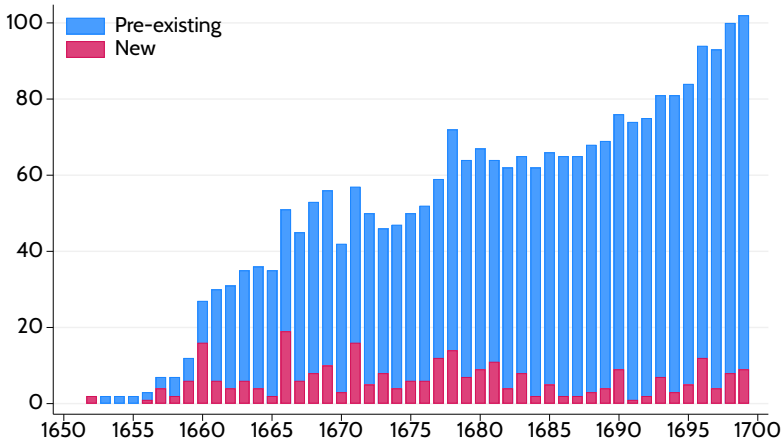
1663 survey lists 82 coffeehouses; we identify 24 in 1663, and 28, 31, 43 within ± 1, 2, 3 years

In the 1690s, administrative records list 124 coffeehouses; we identify 75 over 1692–1698

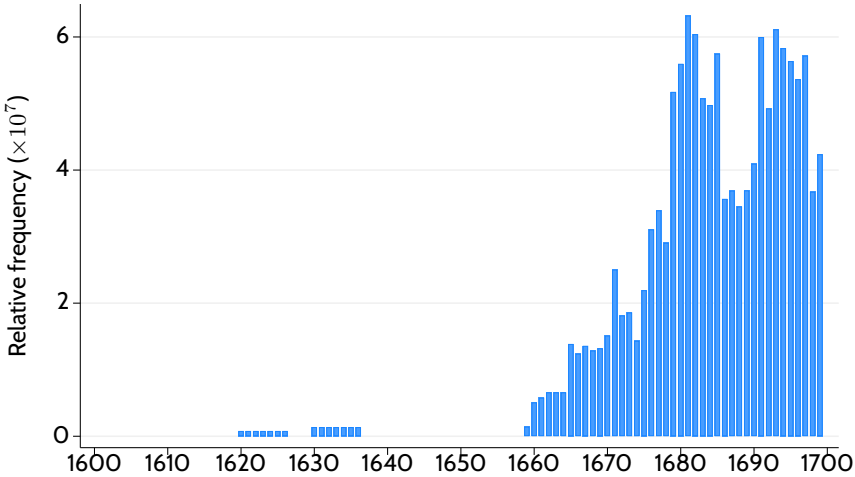
Post-1666 trends in pre-existing and new coffeehouses appear reasonable. Details

Mentions of coffeehouses in print track the series year by year Details

Coffeehouses in London Over Time



Frequency of N-Grams Containing Coffeehouse Terms over Time



Mathematical Practitioner Network

1. Let the set of nodes be our set of mathematical practitioners from Taylor (1954).
2. Link to Warren et al. (2016) to add data on 1/3 of the practitioners on the ODNB.
3. Define

$$g_{ij} \equiv \mathbb{1}\left[\text{nodes } i \text{ and } j \text{ are ever-connected}\right],$$

   where ever-connectedness means having ever been socially or intellectually linked.
4. Define ties over time with

$$g_{ijt} \equiv \mathbb{1}\left[g_{ij} = 1, \text{nodes } i \text{ and } j \text{ are active in period } t\right].$$

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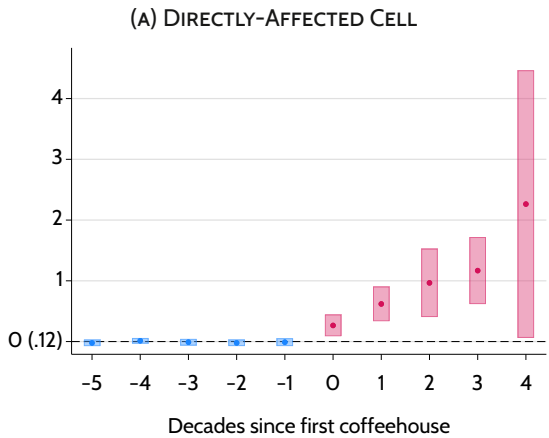
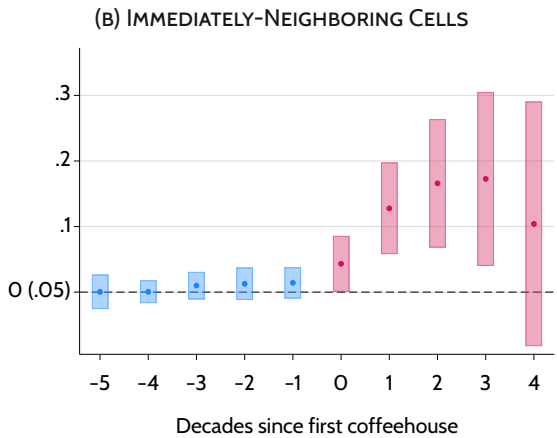
#2 Spillovers Are Inconsistent with Pure Relocation

Scientific Instrument Shops





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The Spillover Design

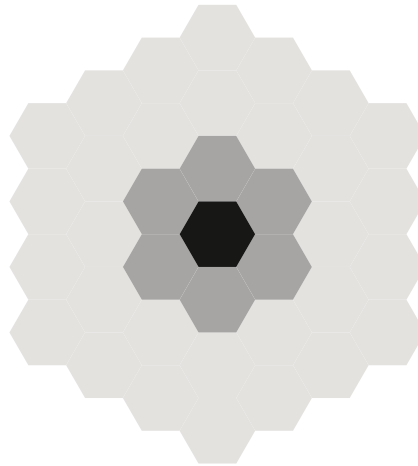
London tiled with **2,830 hexagons**, 200 m across

Opening cell: where a parish's first coffeehouse opens (76 cells)

Neighbors: the six cells around it, 0–200 m

Controls: cells over 1 km from any opening

Two-period DiD (Butts, 2023): one effect for opening cells, one for neighbors, from the 1640s to the 1690s



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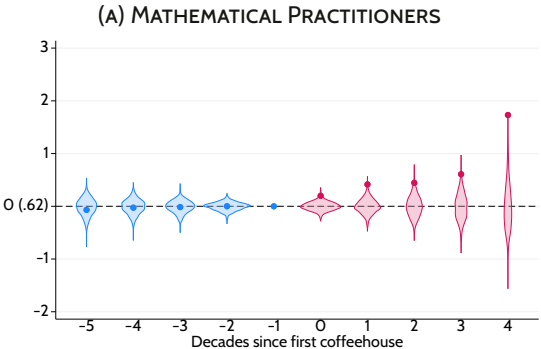
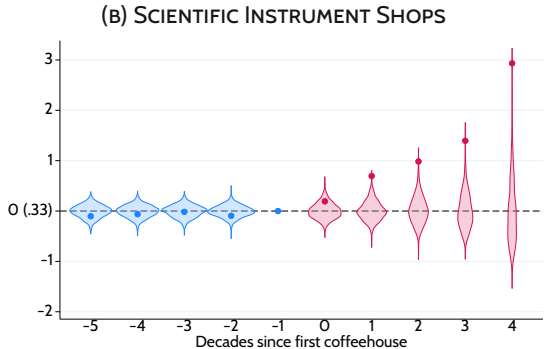
Where Spillovers End: Effects by Distance Ring

		Mathematical practitioners			Scientific instrument shops		
Distance to opening	Hexagons	Mean	No rings	Rings	Mean	No rings	Rings
Opening hexagon	76	0.237	0.381^{**} (0.157)	0.392^{**} (0.159)	0.066	1.182^{***} (0.249)	1.196^{***} (0.250)
0 to 200 m	172	0.081		0.166^{***} (0.054)	0.052		0.121^{**} (0.051)
200 to 400 m	121	0.074		-0.019 (0.048)	0.017		0.048 (0.037)
400 to 600 m	172	0.012		0.009 (0.013)	0.000		0.027^{**} (0.013)
600 to 800 m	225	0.013		0.011 (0.014)	0.004		0.007 (0.011)
800 to 1000 m	158	0.013		0.010 (0.012)	0.000		0.036 (0.027)
Beyond 1,000 m	1906	0.001			0.001		
Hexagons in control group			2754	1906		2754	1906

Change from the 1640s to the 1690s, Conley SEs, 1 km cutoff, 2,830 hexagons · Mean: outcome in the 1640s

Effects on Scientific Activity: Randomization Inference





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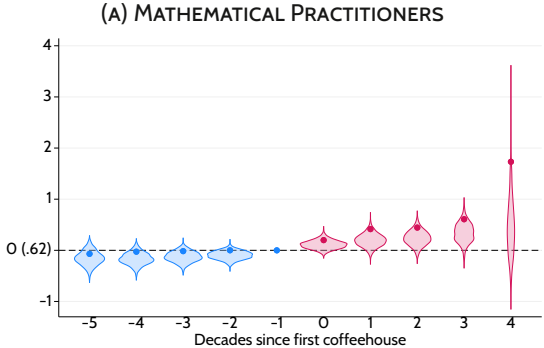
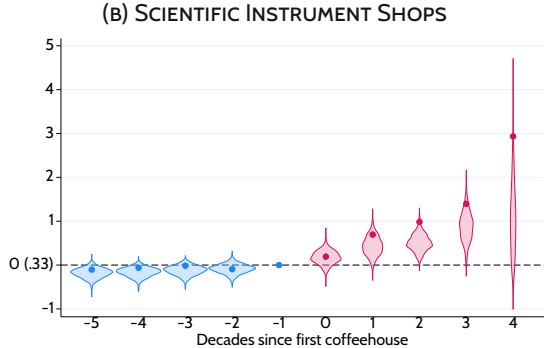
Effects on Scientific Activity: Conditional Randomization Inference





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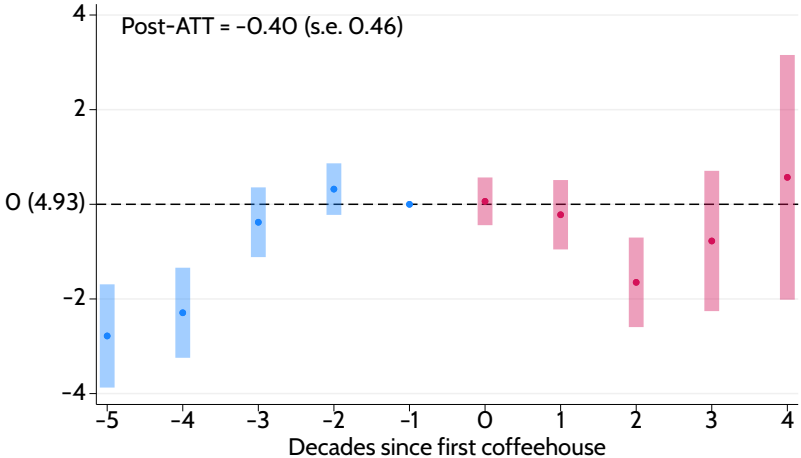
Effects on Placebo Outcomes

Taverns



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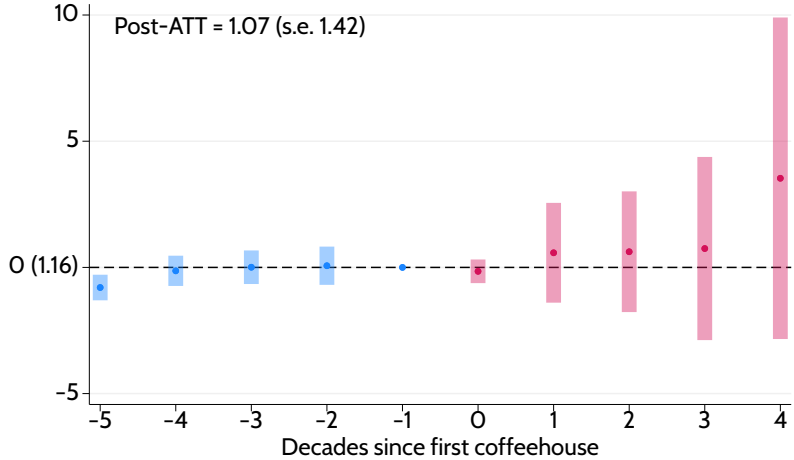
Effects on Placebo Outcomes

Letters Unrelated to Science



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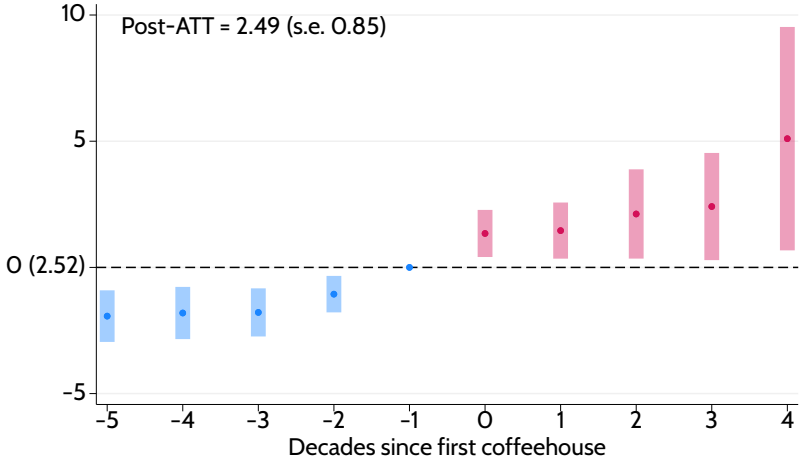
Composition

Effects on Placebo Outcomes

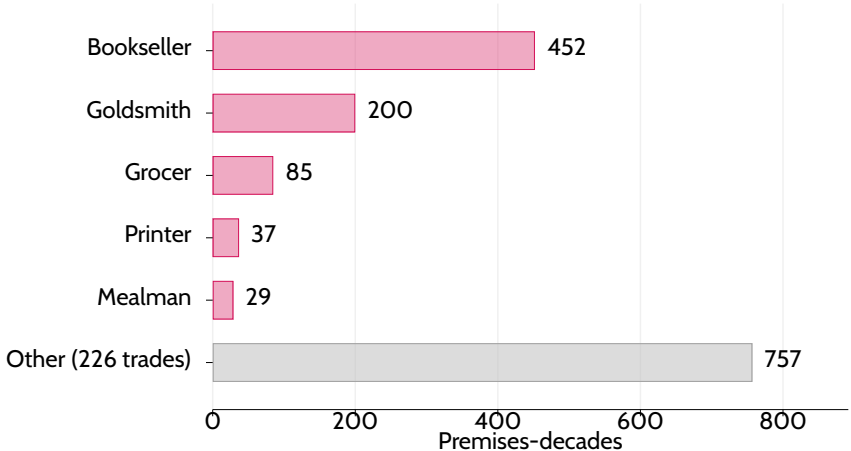
Commercial Premises



Commercial premises exclude taverns, which are examined separately

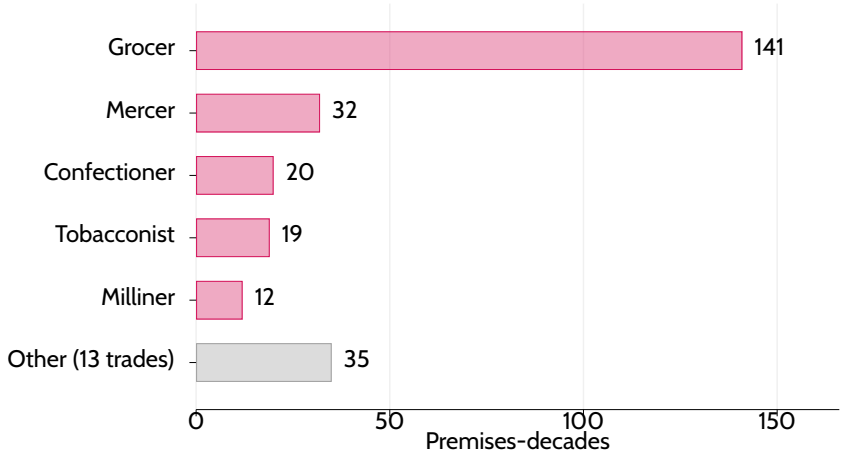
Inside the Placebo Outcomes

Commercial Premises



Inside the Placebo Outcomes

Luxury Shops



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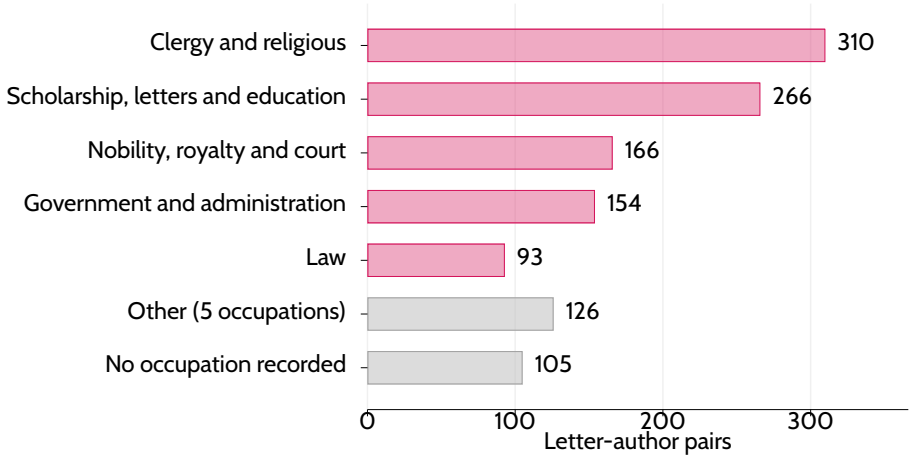
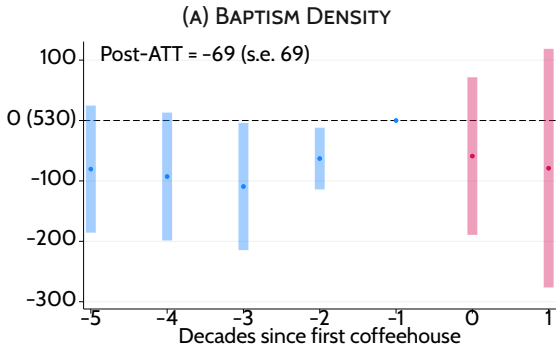
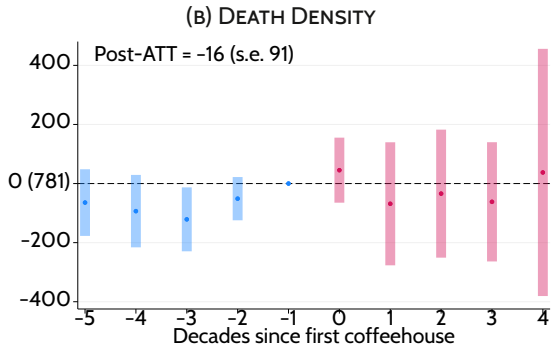
Inside the Placebo Outcomes

Letters Unrelated to Science



Effects on Population Density





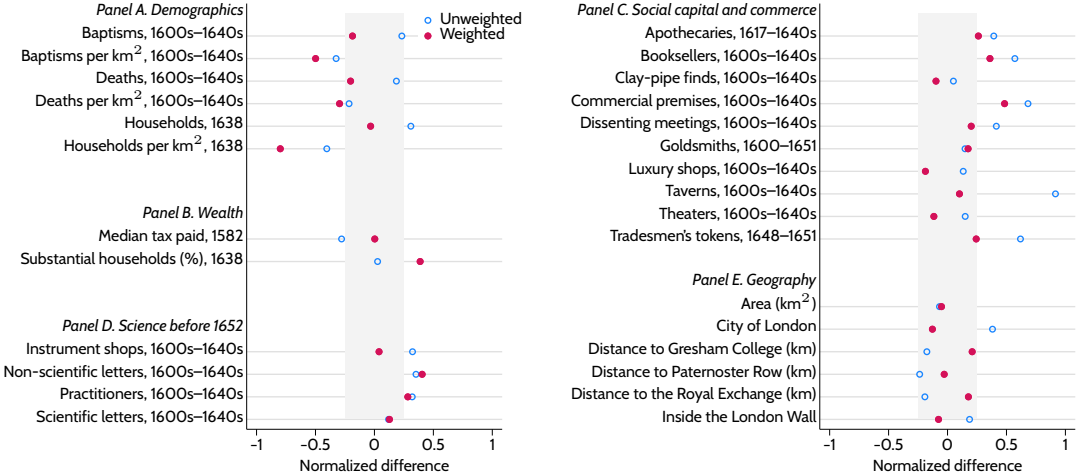
Pre-Treatment Summary Statistics

	Obs.	Mean (SD), 1600s–1640s unless noted: Treatment	Control	Norm. diff.
Panel A. Demographics				
Baptisms	130	84.72 (120.74)	57.44 (113.62)	0.23
Baptisms per km^2	130	493.76 (349.55)	634.63 (503.77)	−0.32
Deaths	110	117.25 (174.38)	86.08 (159.55)	0.19
Deaths per km^2	110	740.21 (564.89)	872.94 (665.37)	−0.22
Households, 1638	107	249.00 (329.67)	157.87 (255.37)	0.31
Households per km^2, 1638	107	3419.34 (1368.22)	4114.09 (2010.83)	−0.40
Panel B. Wealth				
Median tax paid, 1582	98	93.20 (87.91)	124.53 (132.58)	−0.28
Substantial households (%), 1638	92	26.45 (15.99)	25.97 (20.17)	0.03
Panel D. Science before 1652				
Instrument shops	162	0.15 (0.40)	0.05 (0.20)	0.32
Non-scientific letters	162	3.87 (10.41)	1.09 (3.98)	0.35
Practitioners	162	0.40 (0.95)	0.16 (0.48)	0.32
Scientific letters	162	0.38 (1.08)	0.23 (1.38)	0.12

	Obs.	Mean (SD), 1600s–1640s unless noted: Treatment	Control	Norm. diff.
Panel C. Social capital and commerce				
Apothecaries, 1617–1640s	162	0.12 (0.32)	0.03 (0.11)	0.39
Booksellers	162	1.53 (3.29)	0.19 (0.37)	0.57
Clay-pipe finds	162	0.50 (2.50)	0.39 (1.98)	0.05
Dissenting meetings	162	0.33 (0.85)	0.06 (0.34)	0.41
Goldsmiths, 1600–1651	162	0.18 (1.18)	0.05 (0.41)	0.15
Luxury shops	162	0.07 (0.20)	0.05 (0.17)	0.13
Pubs	162	3.01 (3.63)	0.53 (1.21)	0.92
Commercial premises (excl. pubs)	162	0.50 (0.90)	0.06 (0.18)	0.68
Theaters	162	0.08 (0.23)	0.04 (0.19)	0.15
Tradesmen's tokens, 1648–1651	162	1.16 (1.88)	0.23 (1.02)	0.62
Panel E. Geography				
Area (km^2)	162	0.99 (4.23)	1.28 (4.37)	−0.07
City of London	162	0.80 (0.40)	0.63 (0.48)	0.38
Distance to Gresham College (km)	162	1.18 (1.02)	1.37 (1.20)	−0.18
Distance to Paternoster Row (km)	162	1.01 (0.88)	1.24 (1.12)	−0.24
Distance to the Royal Exchange (km)	162	1.01 (1.01)	1.22 (1.21)	−0.19
Inside the London Wall	162	0.66 (0.48)	0.56 (0.50)	0.19
Parishes	162	61	101	

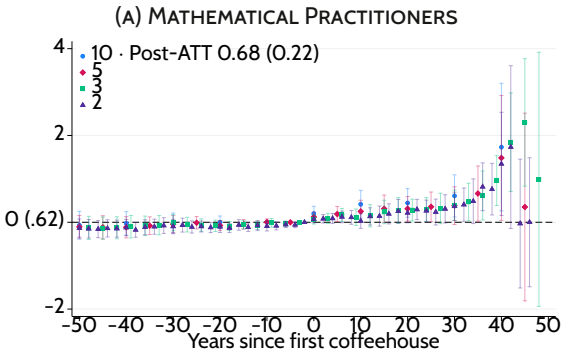
‹ Back | Balance Under IPW

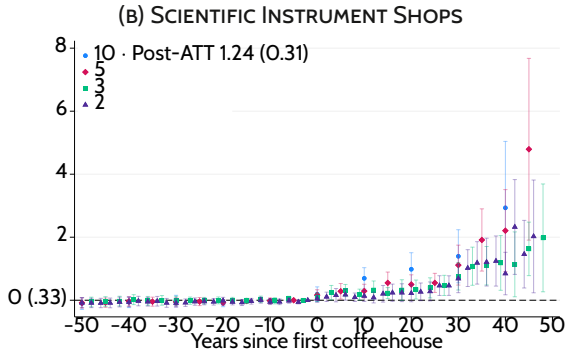
Covariate Balance Before and After Weighting



Score: pre-1652 taverns, tokens, booksellers, City of London

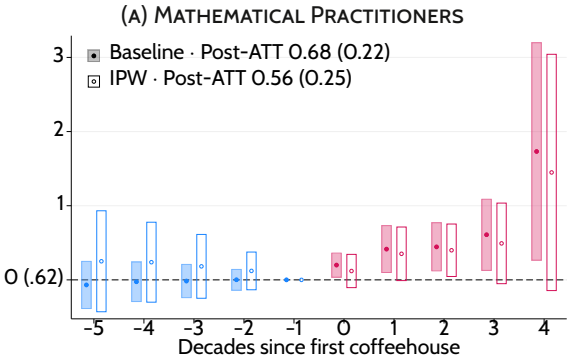
Effects on Scientific Activity by Time-Frame Length

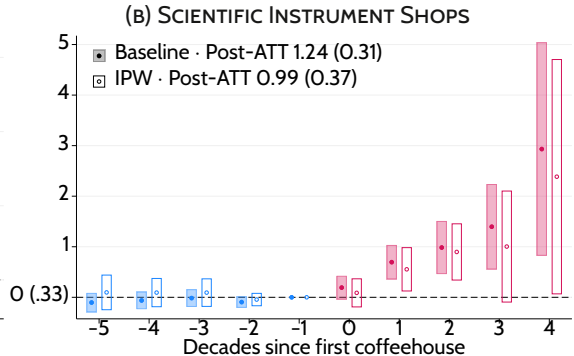




‹ Back

Effects on Scientific Activity: IPW Adjustment

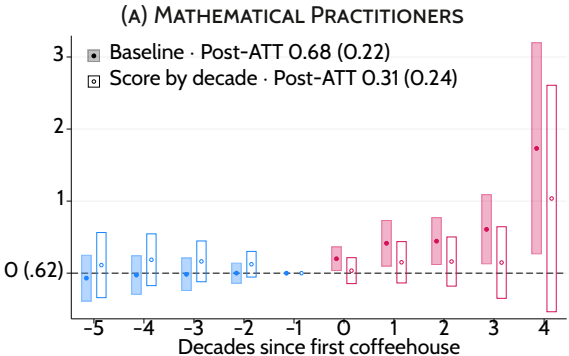


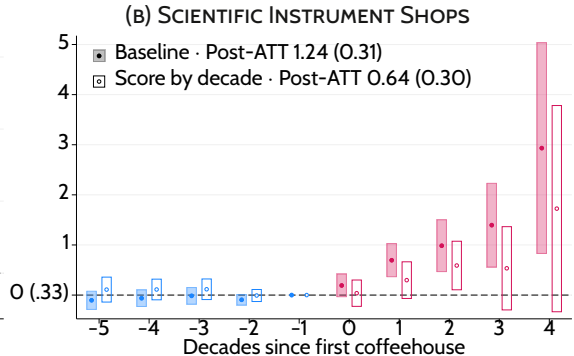


‹ Back

Score: pre-1652 taverns, tokens, booksellers, City of London

Effects on Scientific Activity: Regression Adjustment

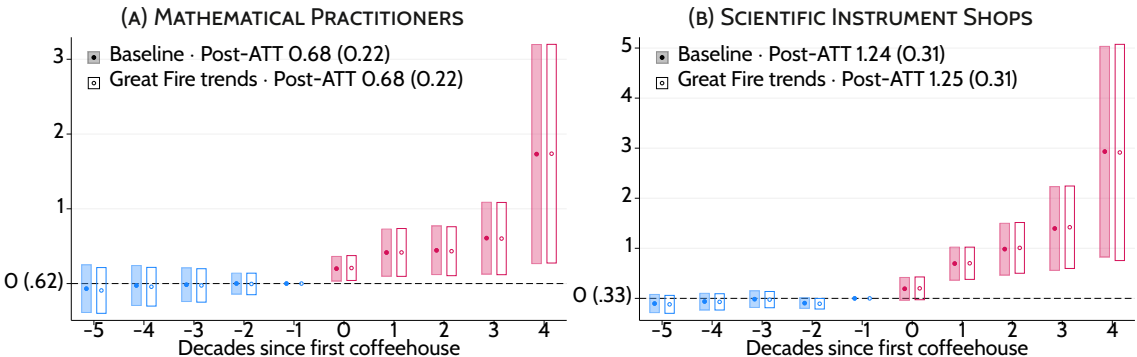




‹ Back

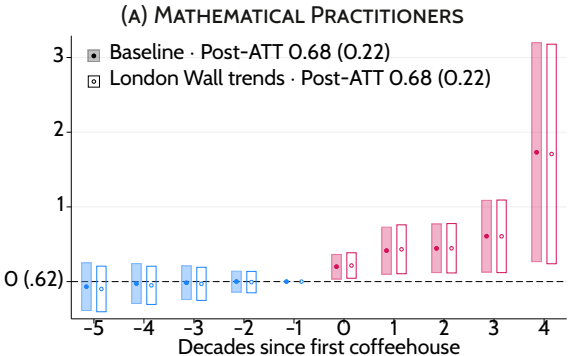
Propensity score × decade FEs · score: pre-1652 taverns, tokens, booksellers, City of London

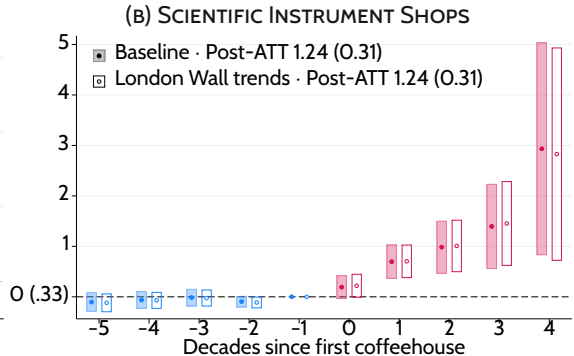
Effects on Scientific Activity: Control for Exposure to the Great Fire



‹ Back

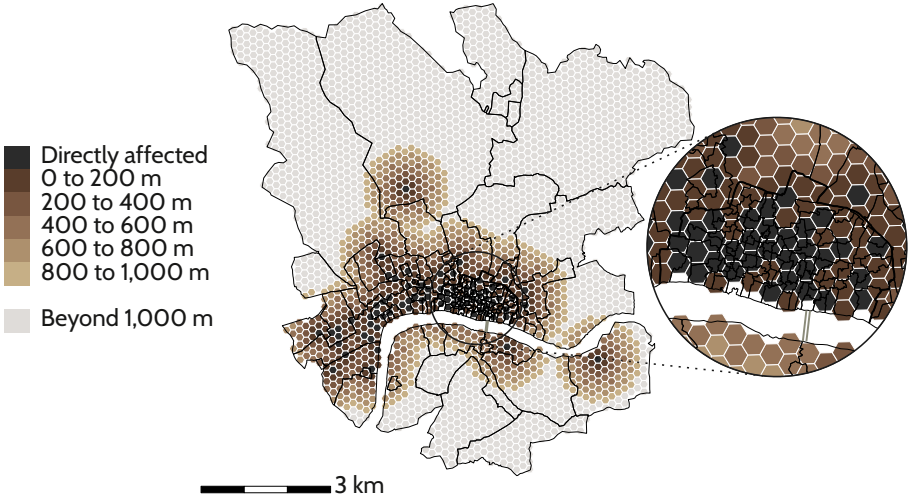
Controlling for Location within the London Wall





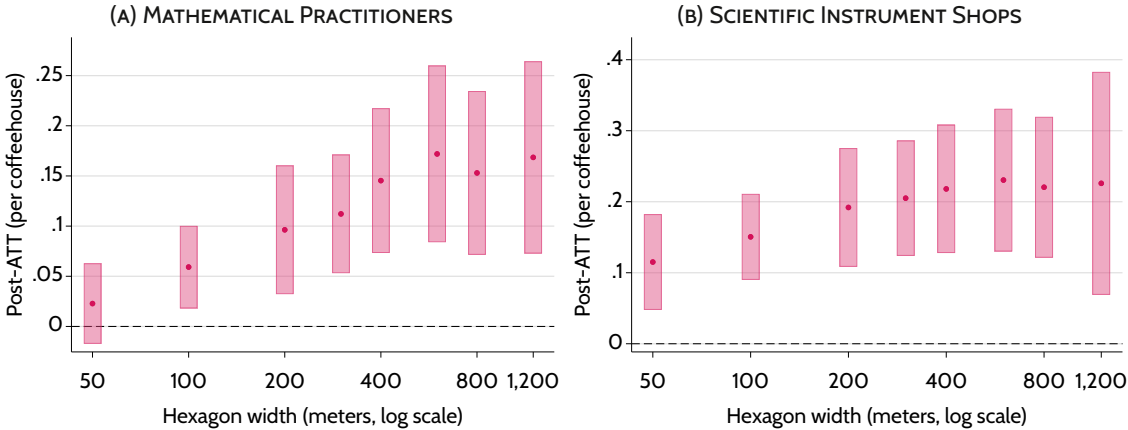
‹ Back

Spillover Rings Around First Coffeehouse Openings



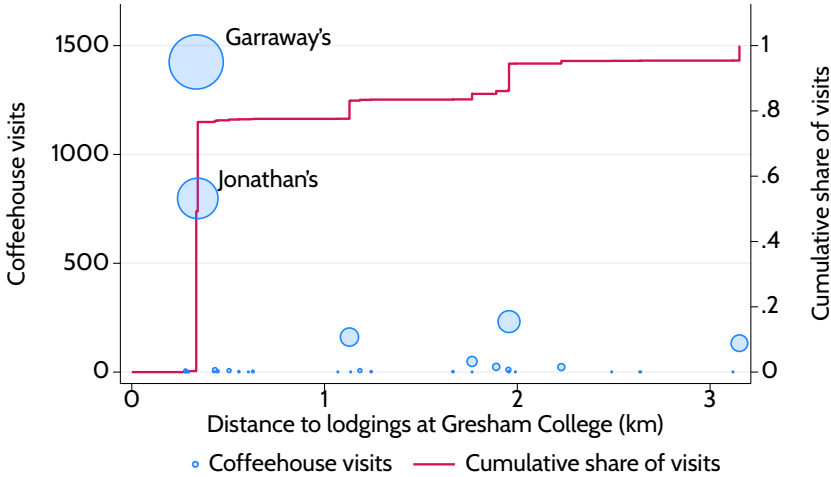
‹ Back

Where Spillovers End: Effects by Hexagon Width

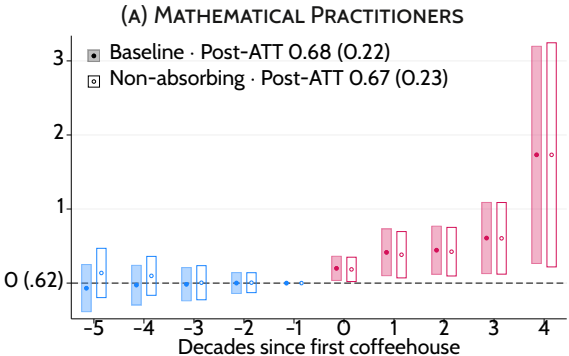


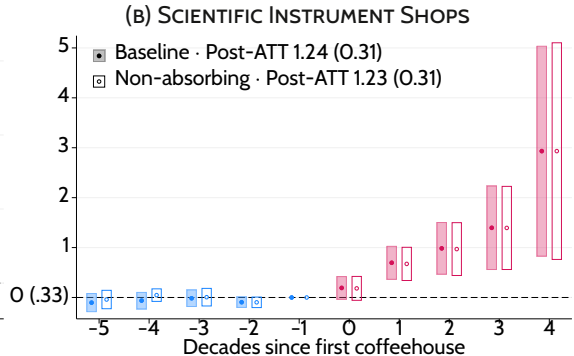
‹ Back | By distance ring | Hooke's coffeehouse visits by distance

Robert Hooke's Coffeehouse Visits, 1672–1683



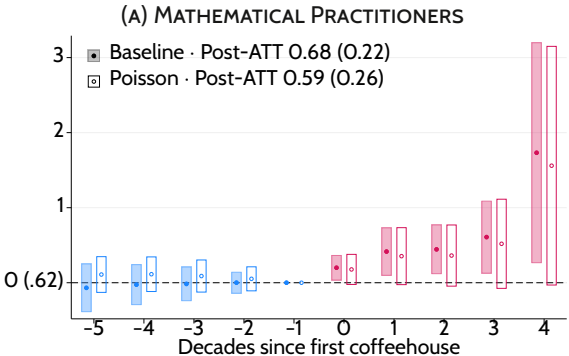
Effects on Scientific Activity: Non-Absorbing Treatment

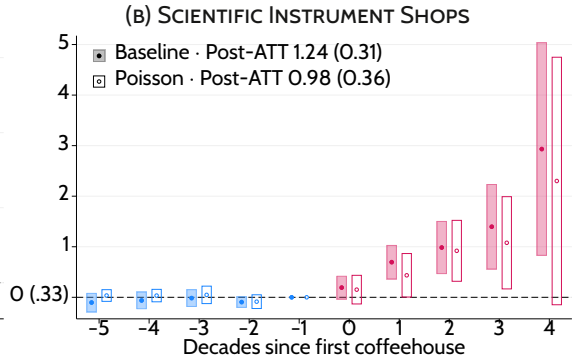




‹ Back

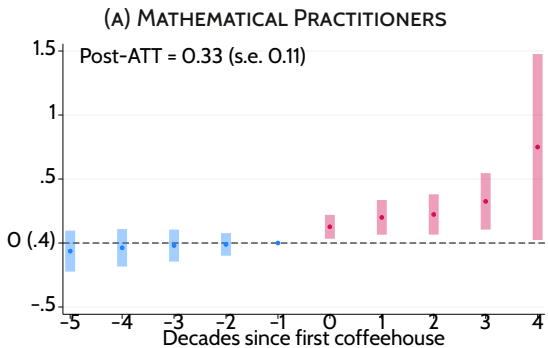
Effects on Scientific Activity: Linear and Poisson

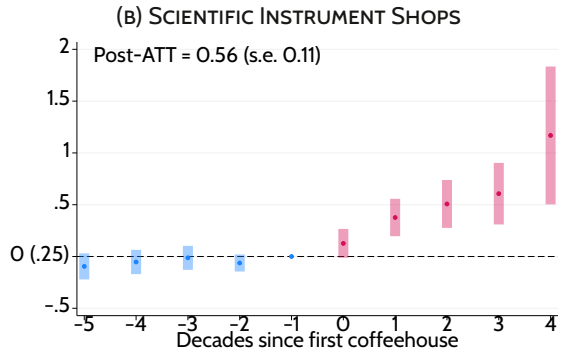




‹ Back

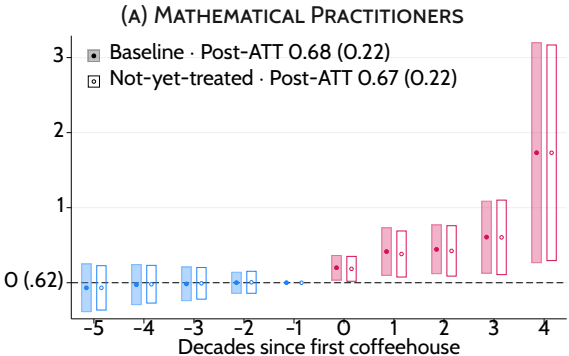
Effects on Scientific Activity: Inverse Hyperbolic Sine

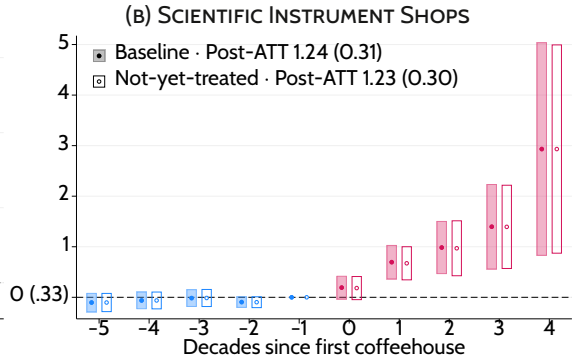




‹ Back

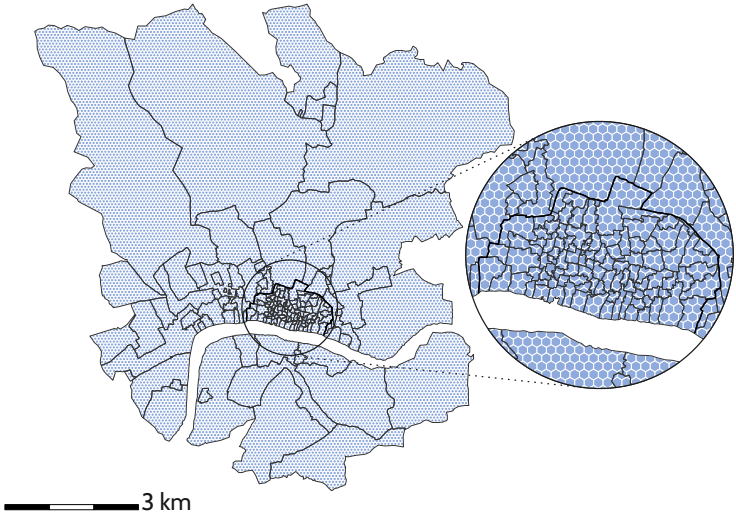
Effects on Scientific Activity: Never- and Not-Yet-Treated



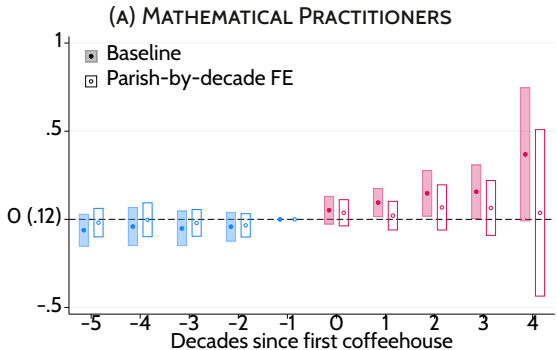


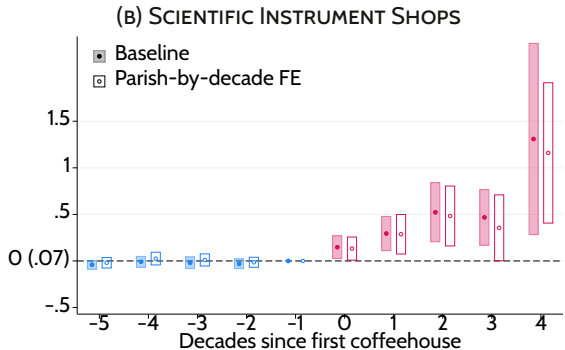
‹ Back

Hexagonal Partition of London



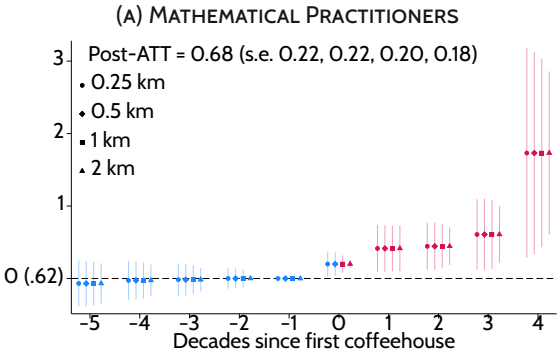
Effects on Scientific Activity: Grid Cell-Level

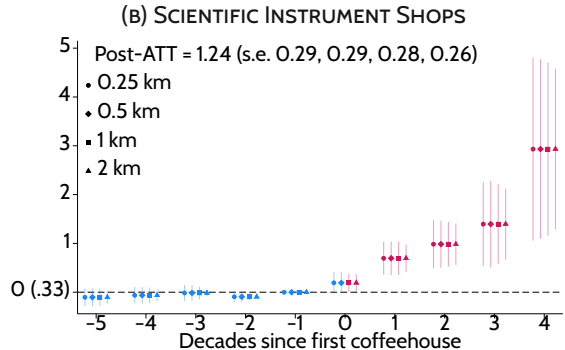




‹ Back

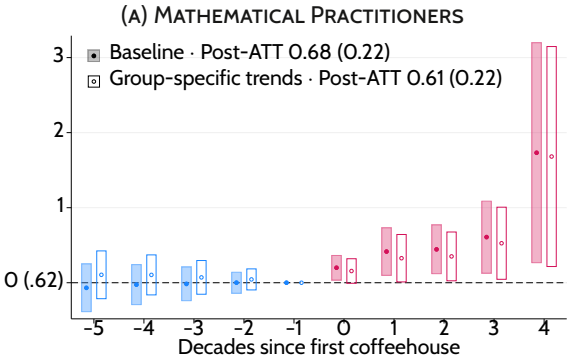
Effects on Scientific Activity: Conley (1999) SEs

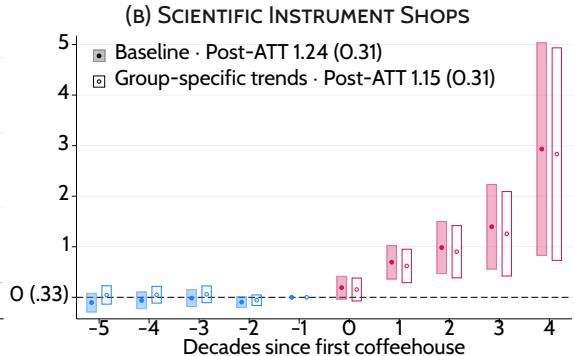




‹ Back

Effects on Scientific Activity: Group-Specific Trends

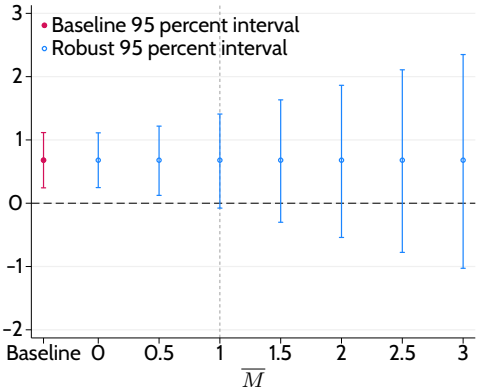




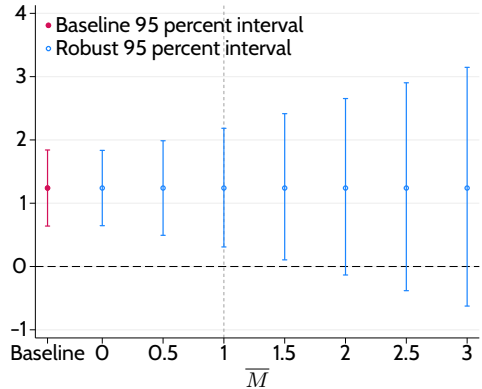
‹ Back

Sensitivity to Violations of Parallel Trends

(a) Mathematical Practitioners



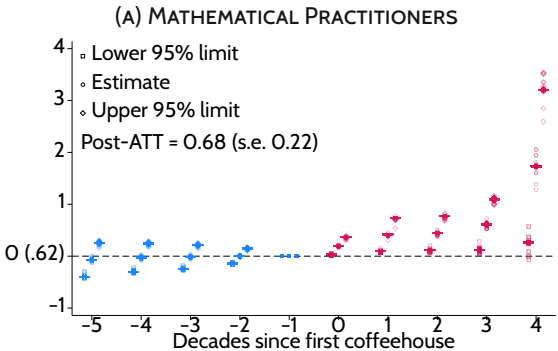
(b) Scientific Instrument Shops

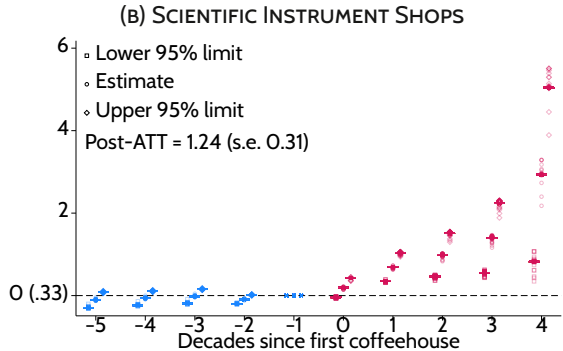


‹ Back

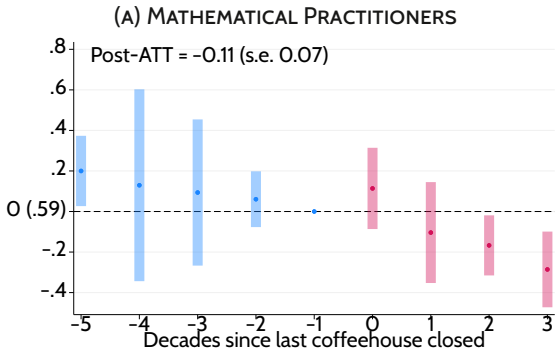
Rambachan and Roth (2023) · significant up to $\overline{M} = 0.8$ for practitioners, 1.7 for shops

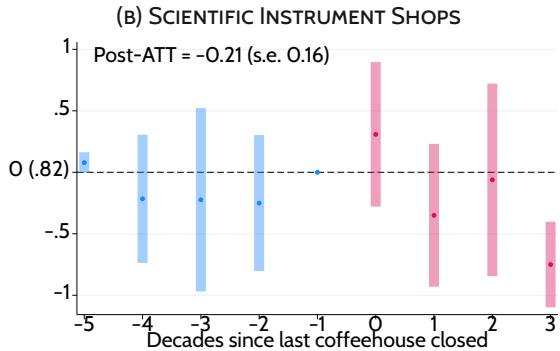
Effects on Scientific Activity: Leave-One-Out Sensitivity Check





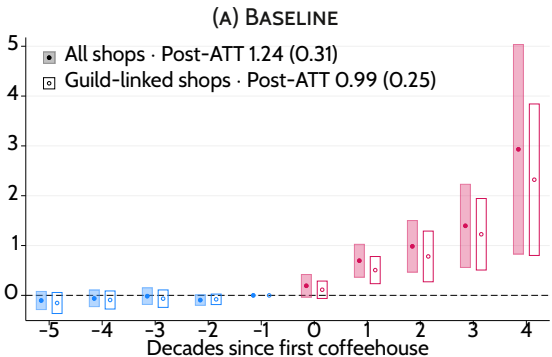
Effects on Scientific Activity: Closing of the Last Coffeehouse

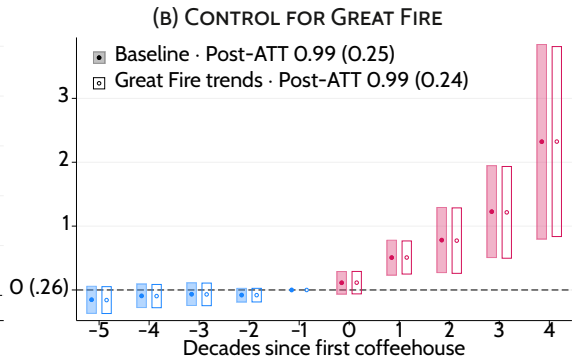




‹ Back

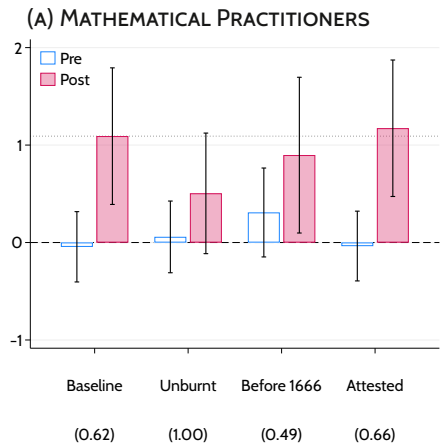
Effects on Scientific Instrument Shops in Guild

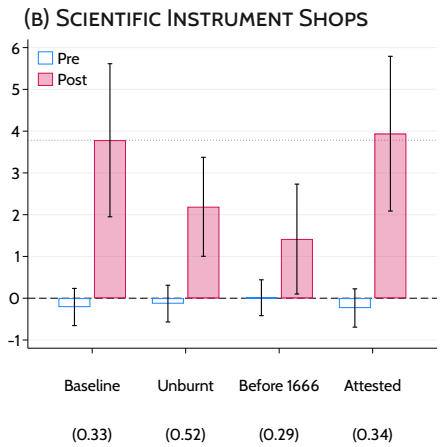




‹ Back

Effects on Scientific Activity: Records Lost in the Great Fire

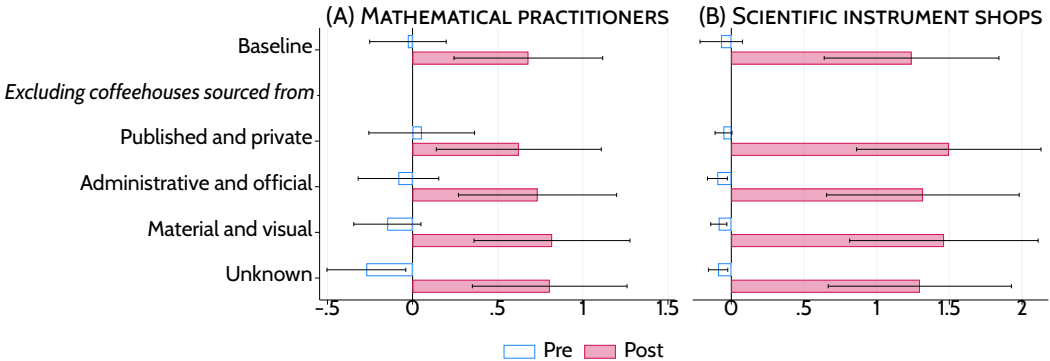




‹ Back

Unburnt: outside the fire area · Before 1666: records to 1666 · Attested: recorded after 1666

Effects on Scientific Activity: Dropping Sources



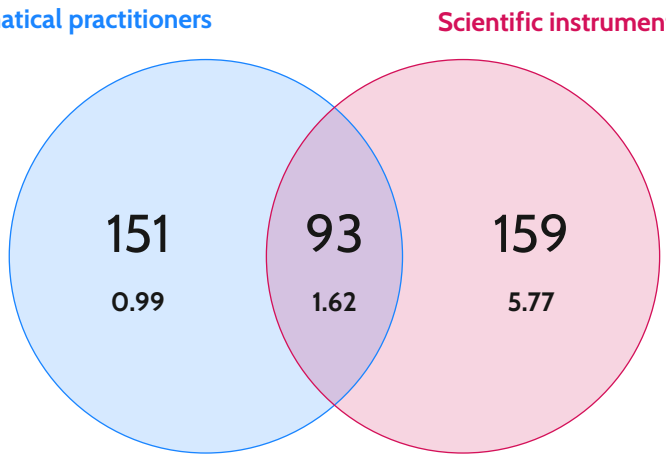
Balance of Pre-Treatment Changes, 1600s–1610s to 1630s–1640s

	Observed parishes	Treatment parishes	Control parishes	Normalized difference
Panel A. Demographics				
Baptisms	130	31.49 (73.83)	17.93 (76.60)	0.18
Baptisms per km^2	130	104.57 (196.59)	107.73 (239.48)	−0.01
Deaths	91	26.44 (100.91)	7.08 (83.22)	0.21
Deaths per km^2	91	10.61 (183.58)	37.26 (286.36)	−0.11
Panel B. Social capital and commerce				
Booksellers	162	−0.04 (2.15)	−0.01 (0.56)	−0.02
Clay-pipe finds	162	0.16 (0.91)	0.26 (1.28)	−0.09
Dissenting meetings	162	0.08 (0.41)	0.03 (0.17)	0.17
Luxury shops	162	0.11 (0.33)	0.09 (0.37)	0.06
Pubs	162	4.16 (4.67)	0.67 (1.63)	1.00
Commercial premises (excl. pubs)	162	0.50 (1.02)	0.10 (0.37)	0.52
Theaters	162	0.02 (0.24)	−0.03 (0.16)	0.25
Panel C. Scientific activity before 1652				
Instrument shops	162	0.09 (0.32)	0.06 (0.36)	0.08
Non-scientific letters	162	0.68 (2.52)	0.24 (1.47)	0.21
Practitioners	162	0.07 (0.81)	0.10 (0.52)	−0.05
Scientific letters	162	0.03 (0.40)	−0.03 (0.68)	0.11
Parishes	162	61	101	

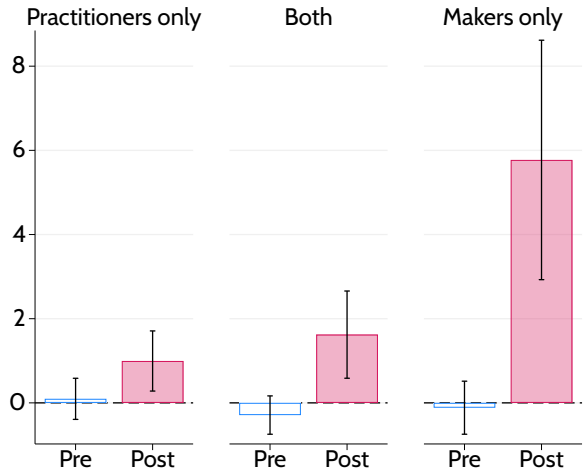
‹ Back

Overlap Between the Outcome Rosters

Post-ATT (relative)

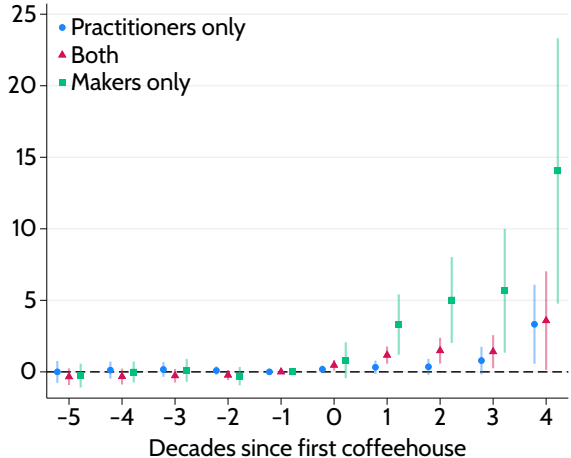


Effects Hold on Each Roster and on Their Overlap



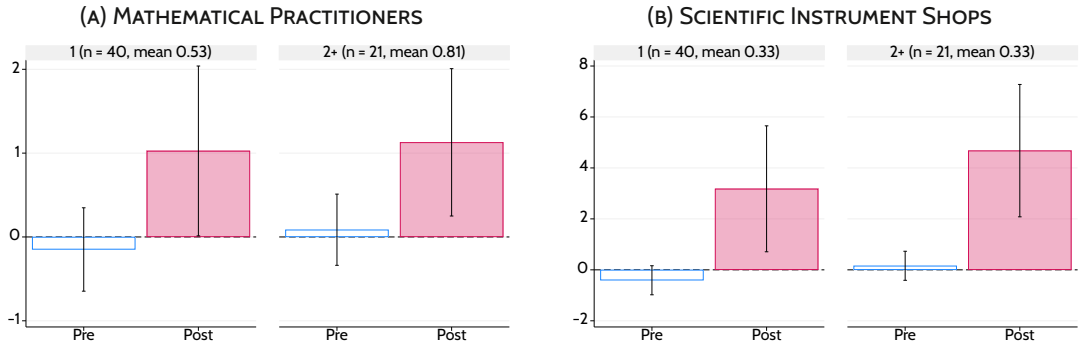
Effects Hold on Each Roster and on Their Overlap

Event Studies by Group

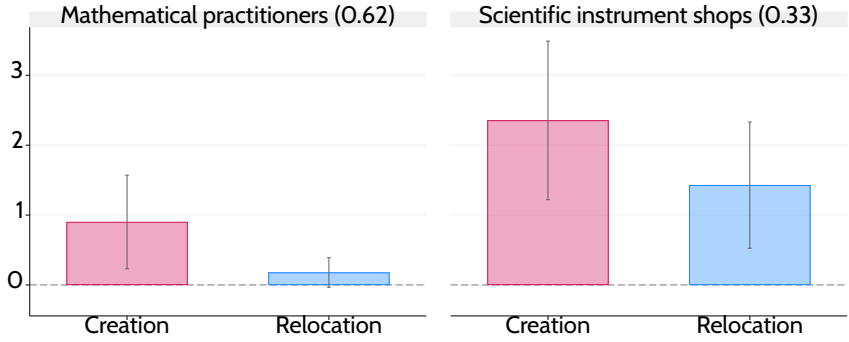


‹ Back

Effects by Coffeehouses at Entry: One versus Two or More

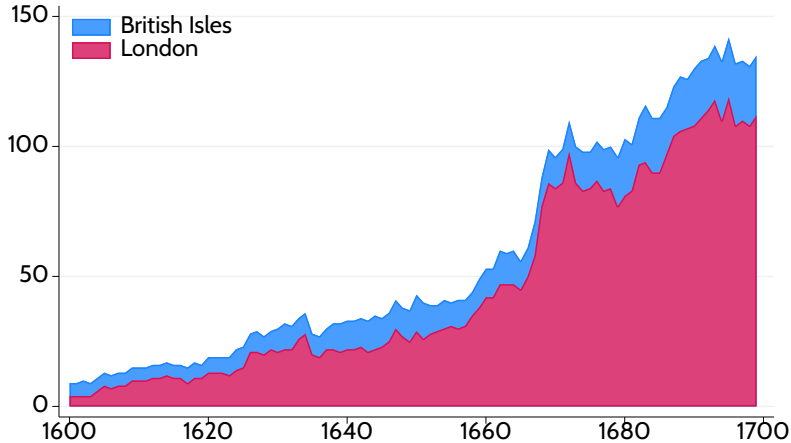


#3 Effects Not Driven by Relocation



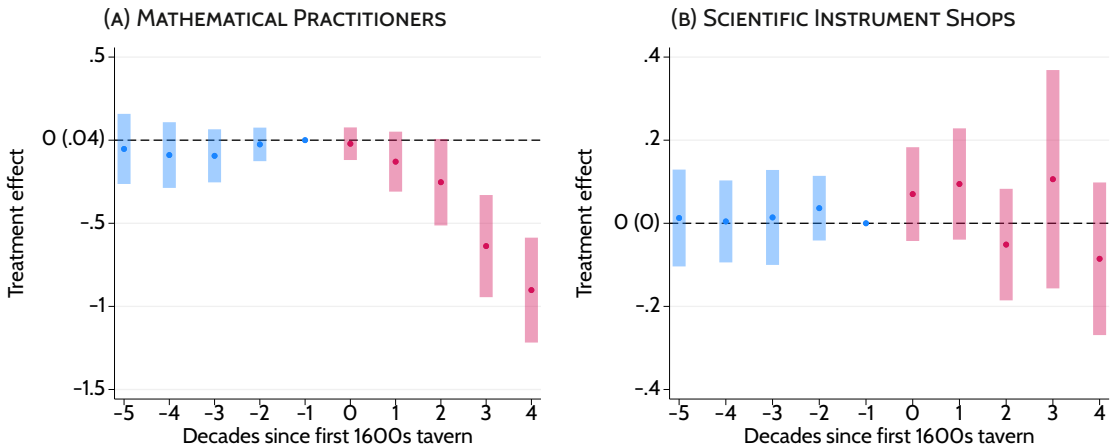
#1 Science Grows in Both London and the British Isles

Scientific Instrument Makers



Taverns Do Not Raise Scientific Activity Before 1650

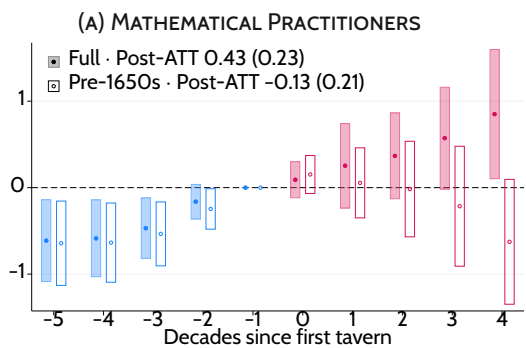
Subtract group-specific trends

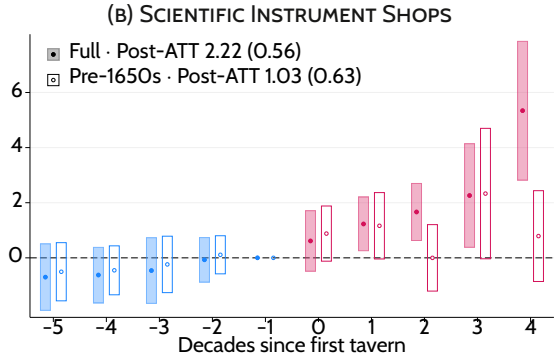


‹ Back | After 1652 | Taverns and coffeehouses after 1652

After 1652, Tavern Openings Track the Expanding Trade

1550s–1690s; over decades 0 to 4, a tavern adds 0.21 shops, a coffeehouse 1.24





‹ Back

Subtract group-specific trends

Tavern and Coffeehouse Openings Are Correlated After 1652

	Coffeehouses (standardized)				Coffeehouses > 0			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Taverns (standardized)	0.408	0.264	0.412	0.251				
	(0.067)	(0.094)	(0.068)	(0.103)				
Taverns > 0					0.169	0.073	0.154	0.017
					(0.025)	(0.035)	(0.024)	(0.032)
Mean dep. var.	0	0	0	0	0.107	0.107	0.107	0.107
Observations	1620	1620	1620	1620	1620	1620	1620	1620
Parish FE		✓		✓		✓		✓
Decade FE			✓	✓			✓	✓

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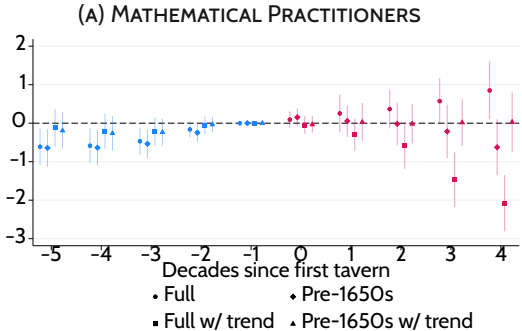
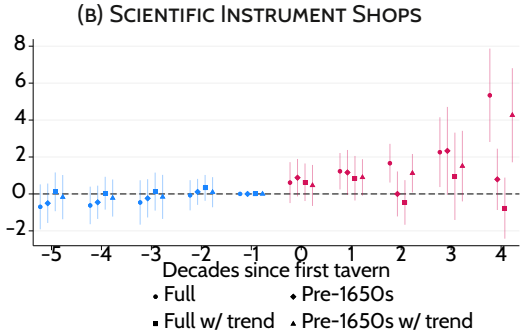
Taverns over the Full Period: With Group-Specific Trends





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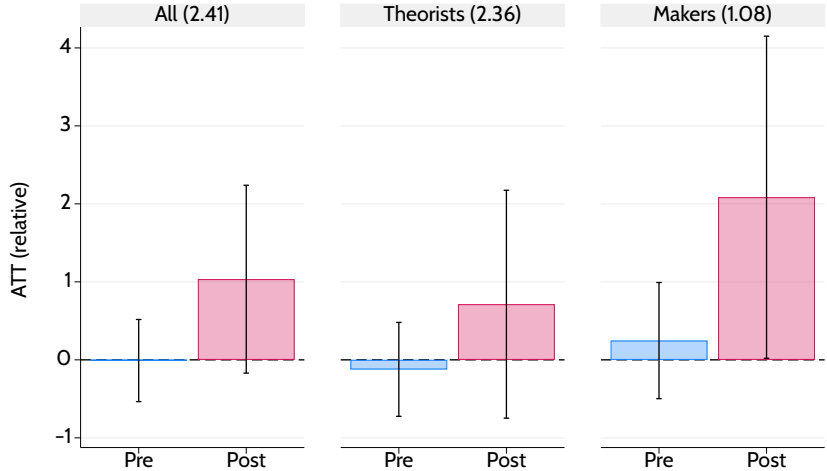
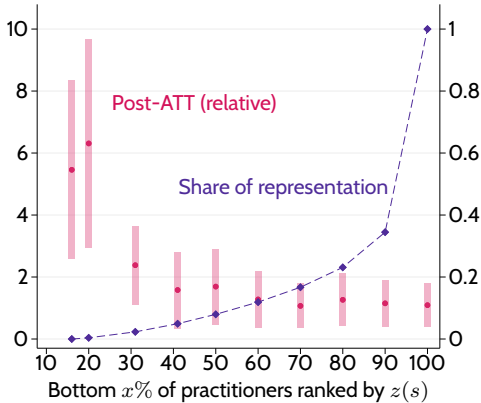
Coffeehouses Connected Makers



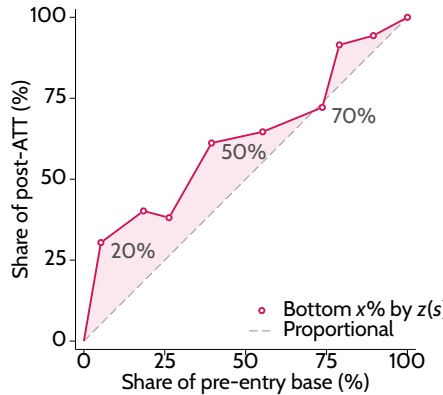
Effects Concentrated Outside Elite Networks

Along the Lorenz Curve

(A) Relative Post-ATT by Bottom x% of $z(s)$



(B) Share of Post-ATT vs. Share of Base



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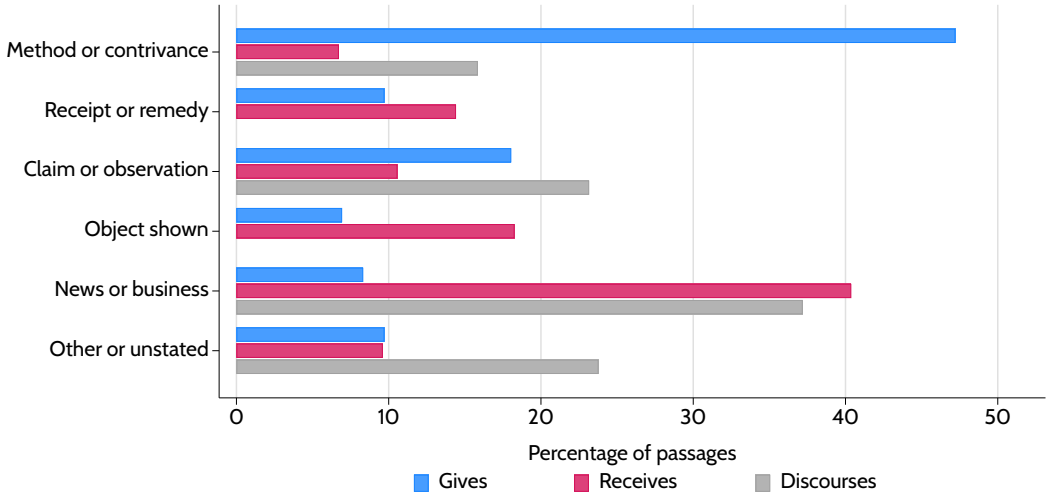
Each point a separate estimate · (b) bottom half by $z(s)$: 39% of the pre-entry base, 61% of the post-ATT

Hooke Gives Methods and Gets News



Three Passages at the Coffeehouse

Gives · 8 March 1675, Garraway's

Tompion. I shewd my way of fixing Double Springs to the inside of the Ballance wheel.

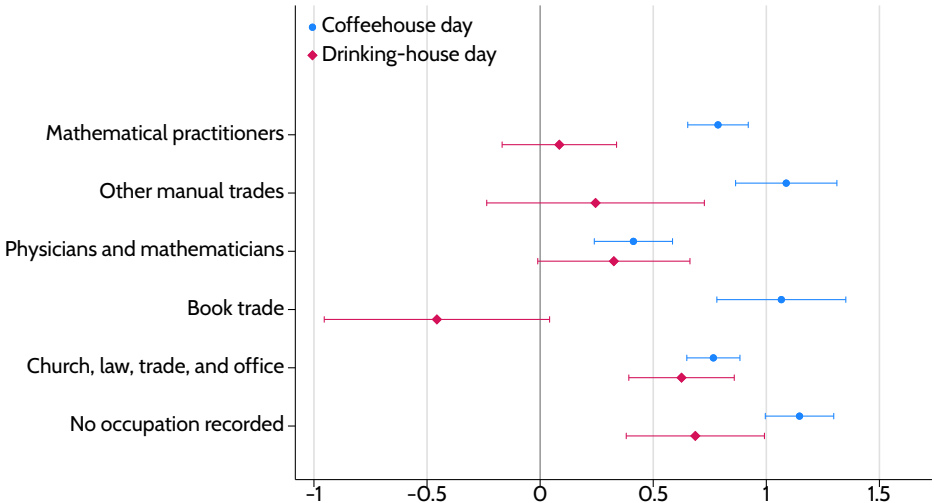
Receives · 7 May 1674, Garraway's

[…] He told me of his way of making a glasse bear a bigger charge by using a peice of reflex mettall. My way much better thus which I told not.

Discourses · 11 December 1675, Joe's

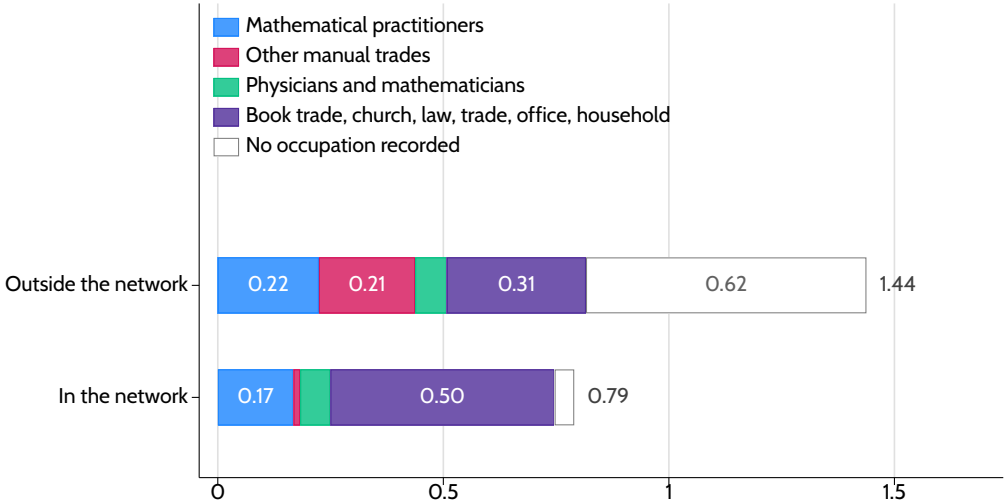
[…] Mr. Aubery discoursd about Mr. Newtons new hypothesis and I declard my way of boyld alabaster […]

Who Coffeehouse Days Bring, and Who Tavern Days Bring



3,464 diary days, 1672–1683 · year × weekday effects

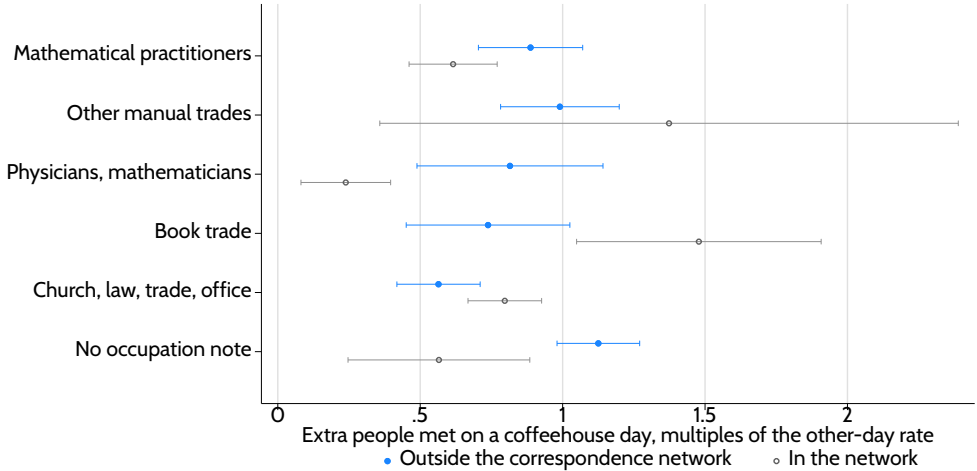
Outside the Network It Is Practitioners and Tradesmen



‹ Back | Each cell against its own rate | 3,464 diary days, 1672–1683 · year × weekday effects

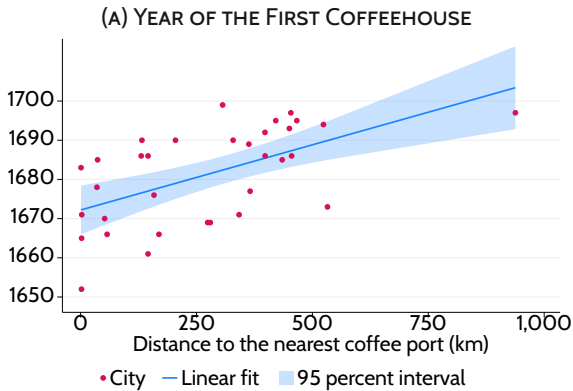
Outside the Network It Is Practitioners and Tradesmen

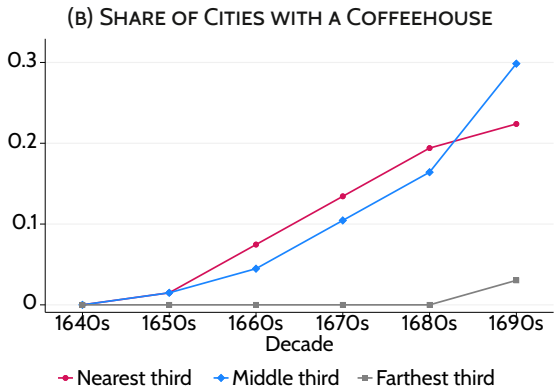
Each Cell as a Multiple of Its Other-Day Rate



3,464 diary days, 1672–1683 · year × weekday effects · in-network manual trades: 8 men on 97 days

Coffee Ports Predict Where and When Coffeehouses Arrived





(a) 33 of 37 cities, 3.3 years per 100 km · ports: Venice, Livorno, Marseille, London, Amsterdam

First Coffeehouses in European Cities

Country	City	Year	Decade	Source
Austria	Vienna	1685	1680s	Albrecht (1989)
Belgium	Antwerp	1690	1690s	Van Aert (2009)
	Kortrijk	1690	1690s	P. D. B. (1934)
Denmark	Copenhagen		1690s	Hoff (2020)
France	Angers	1693	1690s	Archives municipales d'Angers (2010)
	Lyon		1670s	Grand-Carteret (1902); Van Damme (2005)
	Marseille	1671	1670s	Ellis (2004)
	Paris	1671	1670s	Ellis (2004); Leclant (1951)
	Strasbourg	1695	1690s	Perry (2011)
Germany	Bremen	1669	1660s	Ellis (2004)
	Frankfurt am Main	1689	1680s	Cowan (2014)
	Hamburg	1677	1670s	Albrecht (1989); Robilliard (2024)
	Leipzig	1694	1690s	Ellis (2004)
	Lübeck	1695	1690s	Meyer (2011)
	Munich	1699	1690s	Literaturportal Bayern (2019)
	Nuremberg	1686	1680s	Cowan (2014)
	Regensburg	1686	1680s	Cowan (2014)
	Würzburg	1697	1690s	Denzinger (1848)

Country	City	Year	Decade	Source
Italy	Genoa	1661	1660s	Belgrano (1875)
	Rome		1650s	Lombardo (2015–2016)
	Venice	1683	1680s	Ellis (2004)
Netherlands	Amsterdam	1665	1660s	Stadsarchief Amsterdam (2019); Hell (2017)
	Groningen	1686	1680s	Kuiken (1982)
	Leiden	1685	1680s	van Oostrum (2006)
	Middelburg	1686	1680s	City of Middelburg (1686)
	Rotterdam	1666	1660s	Heise (1987)
	The Hague	1670	1670s	van Oostrum (2006); Betz (1900)
	Utrecht	1678	1670s	van Oostrum (2006)
Poland	Gdańsk	by 1697	1690s	Gedanopedia (2020)
Switzerland	Basel		1690s	Meier (1973)
	Geneva	c. 1690	1690s	Encyclopédie de Genève (2020)
UK	Bristol	1666	1660s	Pincus (1995)
	Edinburgh	1673	1670s	Pincus (1995)
	London	1652	1650s	Ellis (2004)
	Newcastle	1692	1690s	Story (1692)
	Norwich	1676	1670s	Withington (2020)
	York	1669	1660s	Pincus (1995)

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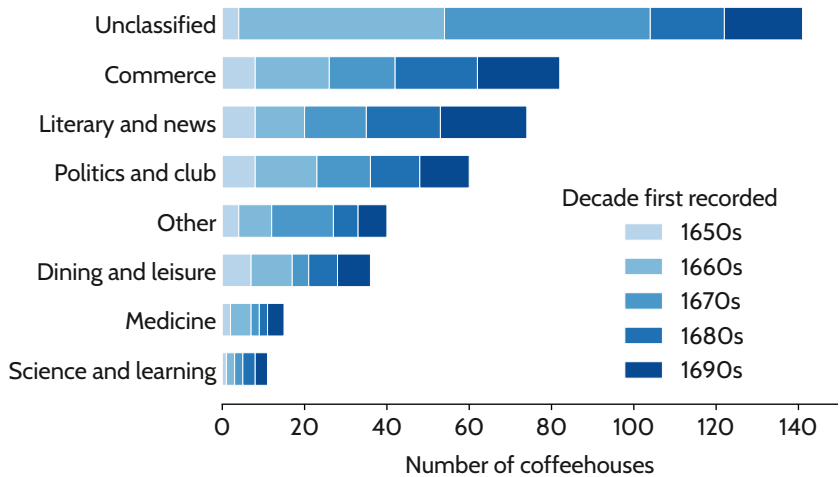
Relative Wealth in the London Victualling Trades, 1692–93

Trade	Individuals	Householders	Mean Rent, £	Mean Stock, £
Victuallers	773	768	21	26
Coffeemen and Women	137	135	30	20
Vintners	131	123	80	149
Innholders or Keepers	91	91	80	74
Tapsters	23	3	—	—
Alehouse-keepers	10	10	21	25
Strongwater-men	10	10	16	42
Total	1,175	1,140		

The Two Cultures of Merchants and Practitioners

Practitioner network size	Practitioners	With commercial tie	Commercial ties / all ties
1–9 ties	7	0	$0/36 \approx 0.0\%$
10–19 ties	15	3	$6/197 \approx 3.0\%$
≥ 20 ties	9	6	$12/348 \approx 3.4\%$
Total	31	9	$18/562 \approx 3.2\%$

What Coffeehouses Were Used For



Coffeehouse Data: Primary Sources

	Any source		Only source	
	Coffeehouses	Per coffeehouse	Coffeehouses	Per coffeehouse
Panel A. Source categories				
Published and private	158	0.55	67	0.23
Printed books, pamphlets, and literary works	47	0.16	9	0.03
Newspapers and periodicals	44	0.15	7	0.02
Secondary histories and reference works	40	0.14	8	0.03
Manuscripts and private papers	36	0.13	13	0.05
Diaries, memoirs, and personal writings	32	0.11	6	0.02
Advertisements, notices, and ephemera	31	0.11	9	0.03
Catalogues, sales, and auctions	8	0.03	2	0.01
Material and visual	98	0.34	44	0.15
Trade tokens	65	0.23	34	0.12
Maps, views, inscriptions, and topographical evidence	36	0.13	10	0.03
Administrative and official	90	0.31	27	0.09
Parish, civic, and local records	44	0.15	13	0.05
Company, commercial, and financial records	34	0.12	7	0.02
State, legal, and court records	33	0.12	6	0.02
Unknown	52	0.18	50	0.17
No source, unidentified, or unclear	52	0.18	50	0.17
Panel B. Summary statistics	Coffeehouses		Per coffeehouse	
Coffeehouses	286			
Sources	617		2.16	
Sources by finer categories	502		1.76	
Sources by broader categories	398		1.39	

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The Model in Equations: Meetings

Meetings. Scientists in parishes p and q, d_{pq} km apart, meet at Poisson rate $\propto e^{\phi D_{pt}} e^{-\delta d_{pq}} e^{\phi D_{qt}}$, with $D_{pt} = 1$ if p has a coffeehouse in decade t and S_{pt} its applied scientists. Documented connections then follow

$$\mathbb{E}[T_{pqt}] = \exp\big(\underbrace{\ln S_{pt} + \ln S_{qt}}_{\text{scientists at each end}} \underbrace{- \delta d_{pq}}_{\text{distance}} + \underbrace{\phi(D_{pt} + D_{qt})}_{\text{coffeehouses}} + \underbrace{\mu_t}_{\text{decade}}\big).$$

Estimated (PPML, connections among the 244 practitioners): $\phi = 0.554$, so a coffeehouse multiplies the rate by $e^{\phi} = 1.74$; $\delta = 0.277$, so the rate halves every 2.5 km.

Access of p, proportional to the meetings each scientist there expects:

$$K_{pt} = \underbrace{e^{\phi D_{pt}}}_{\text{coffeehouse in } p} \sum_q \underbrace{e^{-\delta d_{pq}}}_{\text{distance}} \underbrace{e^{\phi D_{qt}}}_{\text{coffeehouse in } q} \underbrace{S_{qt}}_{\text{scientists in } q}.$$

The Model in Equations: Choices

Choices. Return to practicing in p:

$$R_{pt} = \underbrace{\psi_{pt}}_{\text{fundamental}} \quad \underbrace{K_{pt}^{\eta}}_{\text{access}} \quad \underbrace{S_{pt}^{-\gamma}}_{\text{crowding}}, \qquad \eta, \gamma > 0$$

With Fréchet tastes $\xi_{p\nu}$, $(\zeta_{0\nu}, \zeta_{1\nu})$, potential entrant ν chooses in two steps:

ν practices if $\underbrace{V_t}_{\text{value of practice}} \quad \underbrace{\zeta_{1\nu}}_{\text{taste, science}} > \underbrace{\overline{u}}_{\text{outside trade}} \quad \underbrace{\zeta_{0\nu}}_{\text{taste, trade}}$, $p_\nu = \arg\max_p R_{pt} \quad \underbrace{\xi_{p\nu}}_{\text{taste, parish}}$ if it does,

with $V_t = \left(\sum_k R_{kt}^{\varepsilon_P}\right)^{1/\varepsilon_P}$ what the parishes offer together, and Fréchet shapes ε_P for ξ and ε_S for ζ.

The Model in Equations: Implications

Implications. Where scientists work and how many practice:

$$\underbrace{\ln \sigma_{pt}}_{\text{parish share}} = \underbrace{\beta_P}_{\text{parish margin}} \ln \underbrace{\kappa_{pt}}_{\text{access per London scientist}} + \underbrace{\ln \tilde{\psi}_{pt}}_{\text{fundamental}} + \underbrace{c_t}_{\text{decade}}$$

$$\underbrace{\ln S_t}_{\text{London total}} = \underbrace{\beta_S}_{\text{entry margin}} \ln \underbrace{\overline{K}_t}_{\text{London's average access}} + \underbrace{\bar{c}_t}_{\text{pool}}$$

with

$$\sigma_{pt} = \frac{S_{pt}}{S_t}, \qquad \kappa_{pt} = \frac{K_{pt}}{S_t}, \qquad \overline{K}_t = S_t \Big(\sum_p \tilde{\psi}_{pt} \, \kappa_{pt}^{\beta_P} \Big)^{1/\beta_P}.$$

Calibrated: $\beta_P = 0.56$ to the parish event study; $\beta_S = 0.39$ from European cities, each treated as closed (no moves between cities).

The Model in Equations: A Closed City

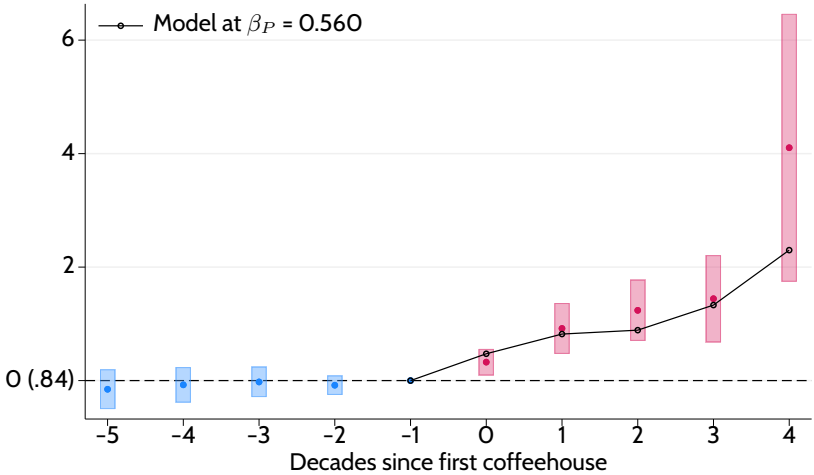
Why a city's first coffeehouse reveals β_S. No scientist moves in or out; the first coffeehouse reaches every parish.

1. Every pair gains ϕ at each end: access per scientist $+2\phi$ (logs) everywhere; shares stay put.
2. Average access, the total times access per scientist: $\Delta \ln \overline{K} = \Delta \ln S + 2\phi$.
3. Entry responds to average access, $\Delta \ln S = \beta_S \, \Delta \ln \overline{K}$, so

$$\underbrace{\Delta \ln S = \frac{2\phi\,\beta_S}{1-\beta_S}}_{\text{model}} = \underbrace{\ln \frac{\text{base}+b}{\text{base}}}_{\text{city event study}}.$$

Titles $5.95 \to 10.92$: $\beta_S = 0.35$. Instrument makers $3.73 \to 8.62$: $\beta_S = 0.43$. Average: 0.39.

The Model Reproduces the Parish Event Study



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Relocation and Creation

Split. A parish's loss from the counterfactual to the observed decade splits into the change in its share at London's average total and its average share of the change in London's total (Bennet, 1920):

$$S_{pt} - S'_{pt} = \underbrace{\overline{S}_t \left(\sigma_{pt} - \sigma'_{pt}\right)}_{\text{relocation: to other parishes}} + \underbrace{\overline{\sigma}_{pt} \left(S_t - S'_t\right)}_{\text{creation: out of science}},$$

with $\overline{S}_t$ and $\overline{\sigma}_{pt}$ the averages of the two decades' totals and shares. Relocation sums to zero over London; creation sums to London's whole loss.

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